

Semisimple Deformations of Contraction Algebras for Simple Curve Flops

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Abstract

Let A be the contraction algebra of a simple flopping curve in a smooth complex threefold, with genus-zero Gopakumar–Vafa invariants n_j . We construct a finite free $\mathbb{C}[[s]]$ -deformation whose geometric generic fibre has n_j matrix blocks of size j , and prove that these multiplicities determine the unique semisimple deformation target. We also prove $\dim HH_0(A) = \dim Z(A) = \sum_j n_j$ and smoothness of A in the scheme of associative multiplications with fixed identity. For existence, a marked ADE morsification separates the curve classes, and the pulled-back tilting generator gives the corresponding matrix blocks. Finiteness and Toda’s dimension formula then establish freeness over the parameter DVR. For uniqueness, we compute a square presentation of the universal marked cocenter whose determinant is the product of the reduced curve-divisor equations. Restriction to the central classifying arc computes its length. Symmetry, stable bimodule periodicity, and Tate duality convert this dimension into an orbit-theoretic uniqueness criterion. The coefficient-five E_8 case is established by an exact modular calculation and a graded tail bound.

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1 Introduction

Let $f : X \rightarrow \operatorname{Spec} R$ be the completion at the contraction point of a simple flopping contraction of a smooth complex threefold, with exceptional curve $C \cong \mathbb{P}^1$. The contraction algebra of Donovan–Wemyss [6] is a finite-dimensional algebra that records noncommutative deformations of $\mathcal{O}_C(-1)$. Let $R \oplus N$ be the basic Van den Bergh generator, with N its nonfree summand. For $\Lambda = \operatorname{End}_R(R \oplus N)^{\operatorname{op}}$ and the free-summand projector e_0 ,

$$A = A_{\operatorname{con}} = \Lambda / \Lambda e_0 \Lambda.$$

We study the semisimple algebras to which A can deform.

The same contraction has genus-zero Gopakumar–Vafa invariants $n_1, \dots, n_\ell$, where $\ell \in \{1, \dots, 6\}$ is its generic length of the scheme-theoretic exceptional fibre and $n_j = 0$ for $j > \ell$. These integers count the isolated $(-1, -1)$ -curves of class $j[C]$ in a generic local deformation [5, 19]. They determine the split semisimple algebra

$$S_{\operatorname{GV}} = \prod_{j=1}^{\ell} M_j(\mathbb{C})^{\times n_j}. \quad (1.1)$$

Toda’s noncommutative-width formula gives

$$\dim_{\mathbb{C}} A = \sum_{j=1}^{\ell} j^2 n_j = \dim_{\mathbb{C}} S_{\operatorname{GV}} \quad [19, \text{Theorem 1.1}].$$

The question is whether this numerical identity is realized by a deformation, and whether the resulting semisimple algebra is uniquely determined.

For finite-dimensional algebras A, B over an algebraically closed field k of characteristic zero, we say that A *deforms to* B if there is a finite free unital $k[[s]]$ -algebra with special fibre A and geometric generic fibre $B \otimes_k \overline{k((s))}$. Jelisiejew–Smoktunowicz prove that this formulation is equivalent to their split polynomial and orbit-closure formulations [12, Theorem 2.20].

Conjecture 1.1 (Wemyss; cf. [12, Questions 1.1–1.2]). *The contraction algebra A deforms to S_{GV} , and every semisimple complex deformation target of A is isomorphic to S_{GV} .*

Hua–Toda describe the matrix factors arising in a generic geometric deformation [10, Theorem 4.2] and formulate a flatness conjecture for contraction algebras and their degree-zero Hochschild (co)homology in geometric families [10, Conjecture 4.3]. Their noncommutative Donaldson–Thomas product formula recovers all GV invariants from the contraction algebra [10,

Theorem 4.4 and Corollary 4.6]. The semisimple-deformation question asks for these invariants to appear as the multiplicities of matrix blocks.

Explicit algebraic constructions were developed by Bowman–Puljić–Smoktunowicz [2, Section 4], who construct a deformation of a nine-dimensional contraction algebra to $M_2(\mathbb{C}) \times \mathbb{C}^5$. Smoktunowicz studies strongly flat deformations and their semisimple specializations in [18]. Using the local normal forms of Brown–Wemyss [4], Jelisiejew–Smoktunowicz prove existence and uniqueness of the semisimple target for Types A and D [12, Theorem 1.3].

We give a uniform construction for simple curve flops of smooth complex threefolds. It yields a finite free formal deformation with the prescribed matrix factors, and a calculation of the marked cocenter proves uniqueness and smoothness of the corresponding algebra law.

Main results

Theorem 1.2 (Semisimple existence). *Let $f : X \rightarrow \operatorname{Spec} R$ be a completed simple flopping contraction of a smooth complex threefold, with contraction algebra A and GV invariants n_j . There is a finite free $\mathbb{C}[[s]]$ -algebra $\mathcal{A}$ such that*

$$\mathcal{A}/(s) \cong A, \quad \mathcal{A} \otimes_{\mathbb{C}[[s]]} K' \cong \prod_{j=1}^{\ell} M_j(K')^{\times n_j}$$

for a finite extension $K'/\mathbb{C}((s))$. After choosing a uniformizer, one may write $K' = \mathbb{C}((u))$ and $s = u^e$.

The proof appears in Section 2. It pulls the universal marked tilting generator through a morsification of the ADE classifying arc. A coefficient- j root contributes $M_j(K')$, and there are n_j such roots. After assembling these factors, a finiteness argument and Toda’s dimension formula prove freeness over the parameter DVR.

For a d -dimensional unital algebra, let X_d denote the affine scheme of associative multiplications on a fixed vector space with fixed identity, and put $G = \operatorname{Stab}_{\operatorname{GL}_d}(1)$.

Theorem 1.3 (Cocenter dimension and semisimple uniqueness). *In the setting of Theorem 1.2, let $d = \dim_{\mathbb{C}} A$. Then*

$$\dim_{\mathbb{C}} HH_0(A) = \dim_{\mathbb{C}} Z(A) = \sum_{j=1}^{\ell} n_j.$$

The point $A \in X_d$ is smooth, and $\overline{G \cdot S_{\operatorname{GV}}}$ is the unique irreducible component through A . Every finite free formal deformation of A with geometrically semisimple generic fibre has geometric Wedderburn type S_{GV} .

Here $HH_0(A) = A/[A, A]$ is the Hochschild cocenter. Section 4 computes it from the universal marked cocenter over $R_H = \widehat{\mathcal{O}}_{H,0}$, where $H = \mathfrak{h}/W_0$ is the marked ADE base of Section 2. A square presentation matrix Θ of this module has determinant equal, up to a unit, to $P_1 \cdots P_{\ell}$, with $P_j = 0$ the reduced divisor of coefficient- j curves. Restricting to the original classifying arc f_0 gives

$$\dim_{\mathbb{C}} HH_0(A) = \operatorname{ord}_w \det \Theta(f_0(w)) = \sum_j \operatorname{ord}_w P_j(f_0(w)) = \sum_j n_j.$$

Symmetry identifies the centre with the dual cocenter. Stable bimodule periodicity and Tate duality then give $\dim HH^1(A) = \dim HH^2(A)$, reducing the fixed-unit tangent dimension to $\dim(G \cdot S_{GV})$.

The smoothness statement proves the assertion of [12, Conjecture 1.4(1)] in the smooth simple-flop setting. The orbit-dimension argument discussed in [12, Section 1 and Theorem 2.14] gives the last step: a semisimple orbit whose closure contains A has the same dimension as $T_A X_d$, so the local ring at A is regular. Section 3 presents the fixed-unit form of this criterion.

The marked geometry and the computation

The simultaneous-resolution theory of Katz–Morrison and Bryan–Katz–Leung describes a simple flop by an arc in a marked ADE base [15, 5]. A small translation separates its intersections with the curve divisors into transverse roots. The resulting ordinary double points carry the local matrix factors in the existence proof (Figure 1). The degree of the extended determinant line bundle identifies the size of each factor.

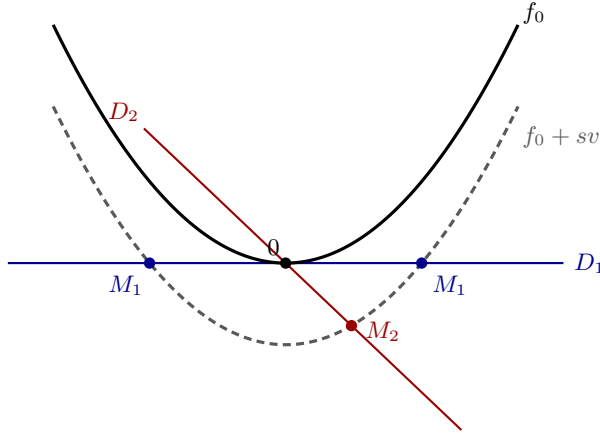


Figure 1: A schematic two-divisor slice of the marked base. A small translation separates a tangency with D_1 into transverse intersections; an intersection with D_j contributes the matrix block M_j . The counts for a given flop are the conserved intersection numbers of its classifying arc.

Etingof–Rains construct a finite free central extension of the preprojective algebra [8]. We prove that the cocenter of its marked corner is free and compute the determinant of multiplication by the central element. The nodal calculations of Etingof–Latour–Rains provide the required closed-fibre dimensions for lengths 2, 3, 4, 6, and length one is immediate [9]. We establish the remaining coefficient-five E_8 case by an exact modular rank calculation together with a grading bound. Its presentation and the numerical data needed for the proof are given in Section 4.5 and Appendix B.

The computational material is available in the companion repository at commit [5a2001f](#). The constructions, verification procedures, and rank tables are described in Appendices A and B.

Conventions and organization. All algebras are associative and unital. The geometric arguments are over $\mathbb{C}$; Section 3 works over an algebraically closed characteristic-zero field. We

write $A^e = A \otimes A^{\text{op}}$, $HH^i(A) = \text{Ext}_{A^e}^i(A, A)$, and $D = \text{Hom}_{\mathbb{C}}(-, \mathbb{C})$ in the complex setting. The notation $[A, A]$ always means the linear span of commutators; the commutative abelianization is the quotient by the ideal they generate. In the geometric construction, w is the surface-slice parameter and s is its external deformation parameter. In Section 4.4, z denotes the central element of the Etingof–Rains extension.

Section 2 constructs the finite free deformation. Section 3 proves the orbit criterion. Section 4 computes the marked cocenter and proves uniqueness. The appendices give the root calculations and computational data.

2 Semisimple existence

Throughout this section the ground field is $\mathbb{C}$. Let

$$f: X \longrightarrow Y = \text{Spec } R$$

be the completion at the contraction point of a simple flopping contraction of a smooth complex threefold. Thus $\text{Exc}(f) = C \cong \mathbb{P}^1$, the scheme-theoretic length ℓ belongs to $\{1, \dots, 6\}$, and the contraction algebra is denoted by A_{con} . We write n_j for the genus-zero Gopakumar–Vafa invariant in class $j[C]$.

Theorem 2.1 (Semisimple existence). *Let f be any completed simple one-curve flop of a smooth complex threefold, of length ℓ , with contraction algebra A_{con} and genus-zero GV invariants n_j . There exists a finite-free $\mathbb{C}[[s]]$ -algebra $\mathcal{A}$ such that*

$$\mathcal{A}/(s) \cong A_{\text{con}}.$$

After a finite splitting extension $K'/\mathbb{C}((s))$,

$$\mathcal{A} \otimes_{\mathbb{C}[[s]]} K' \cong \prod_{j=1}^{\ell} M_j(K')^{\times n_j}. \quad (2.1)$$

The proof has three parts. First we construct a formal morsification and prove that it contains exactly n_j transverse roots of coefficient j . Second we prove that the universal marked tilting bundle and its endomorphism algebra survive the generally nonflat classifying base change, and that a coefficient- j root contributes the complete matrix block M_j . Finally we assemble the local blocks, exclude infinitesimal thickening, and deduce finite freeness from Toda’s dimension formula.

Definition 2.2 (Marked partial-resolution data). Let Φ be the primitive ADE root system of a general hyperplane section, with complex Cartan space $\mathfrak{h}$, Weyl group W , and simple roots $\Delta = \{\alpha_i\}$, determining the positive roots Φ^+ . The retained vertex is α_* . A marking fixes this vertex; on an endomorphism algebra, it fixes the corresponding summand projector. For a root $\alpha = \sum_i c_i(\alpha)\alpha_i$, its marked coefficient is $c_*(\alpha)$; the Levi roots are those with $c_*(\alpha) = 0$. Put

$$W_0 = \langle s_i : i \neq * \rangle, \quad H = \mathfrak{h}/W_0,$$

and let $\pi_W : H \rightarrow \mathfrak{h}/W$ be the induced finite map to the semiuniversal ADE base. Choose affine coordinates on H centred at the origin.

If θ is the highest root, the length classification gives

$$\ell = c_*(\theta)$$

[15]. The curve divisors and symbolic thickenings are indexed by $1 \leq j \leq \ell$ [5, Proposition 2.3 and Lemma 2.4]. Modulo the Levi root lattice, $[\alpha] = c_*(\alpha)[\alpha_*]$. The comparison with curve classes is expressed by degrees of extended line bundles in Lemma 2.8.

The six possibilities are shown in Figure 2.

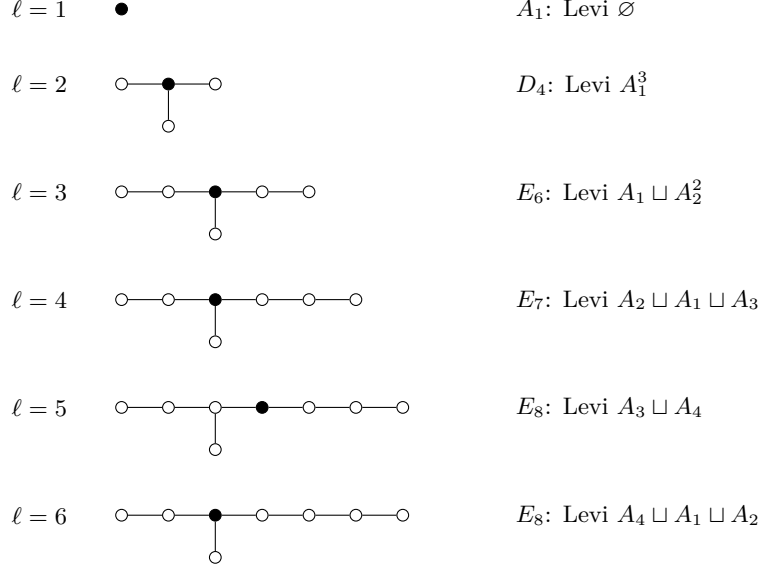


Figure 2: The six primitive marked ADE pairs. The filled vertex is the retained vertex α_* ; $\ell = c_*(\theta)$ is its coefficient in the highest root, and the Levi type is the diagram left after deleting it. For lengths 2, 3, 4, 6 the retained vertex is the branching vertex, which is the case treated by Etingof–Latour–Rains; length 5 is the one exception, and is the case settled by the computation of Proposition 4.9.

Lemma 2.3 (The marked analytic classifying arc). *After replacing the source and target of the contraction by sufficiently small analytic neighbourhoods, there is a holomorphic map*

$$f_0: \mathbb{D}_w \longrightarrow H, \quad f_0(0) = 0,$$

for which the given analytic contraction is isomorphic, as a marked contraction, to the pullback of the simultaneous partial resolution over H . Its completion is a formal classifying arc

$$\nu_0: \mathrm{Spf} \mathbb{C}[[w]] \longrightarrow H,$$

whose pullback is isomorphic to the original completed marked contraction.

Proof. Choose a general analytic hyperplane parameter w through the contraction point. Its central fibre is the marked partial resolution of the ADE surface section. The semiuniversal deformation spaces of the full resolution, the marked partial resolution, and the ADE surface are, respectively,

$$\mathfrak{h}, \quad H = \mathfrak{h}/W_0, \quad \mathfrak{h}/W.$$

Bryan–Katz–Leung construct compatible analytic classifying maps from the w -disk and identify the resolved threefold neighbourhood with the Cartesian pullback of the simultaneous marked partial resolution [5, Section 2.1, especially Propositions 2.2–2.3]. Completing this Cartesian diagram at $w = 0$ gives the required isomorphism of marked formal contractions. $\square$

Let S_0 be the central partial-resolution surface, and write $\mathcal{I}_C \subset \mathcal{O}_{S_0}$ for the ideal of C in that surface.

Proposition 2.4 (The marked family and GV multiplicities). *Applied to the classifying arc of Lemma 2.3, the simultaneous partial resolution has the following properties.*

- (i) *The base H is smooth affine space, the quotient $\pi_W: H \rightarrow \mathfrak{h}/W$ is finite, and the universal simultaneous partial resolution*

$$\pi_\ell: \mathcal{Y}_\ell \longrightarrow \mathcal{X}_\ell = \operatorname{Spec} \mathcal{R}_\ell$$

is projective with source and target flat over H . Its fibres have dimension at most one and

$$R\pi_{\ell*} \mathcal{O}_{\mathcal{Y}_\ell} \cong \mathcal{O}_{\mathcal{X}_\ell}.$$

The universal flopping algebra is finite over its centre $\mathcal{R}_\ell$.

- (ii) *The curve discriminant is a union of precisely ℓ reduced irreducible divisors*

$$D^{\text{curv}} = D_1 \cup \cdots \cup D_\ell.$$

At a general point of D_j , the partial-resolution surface contains one smooth (-2) -curve in class $j[C]$. These curves form a projective flat family $C_j \rightarrow D_j$, and its central fibre is the symbolic thickening C_j defined by $\mathcal{I}_C^{(j)}$.

- (iii) *A sufficiently small boundary-proper generic perturbation of f_0 meets D_j in k_j points, counted with multiplicity, where k_j is the length of the local Hilbert scheme at C_j .*

- (iv) *Toda’s pure thickening is*

$$\mathcal{O}_{C^{(j)}} = (\mathcal{O}_{S_0}/\mathcal{I}_C^{(j)})/Q,$$

where Q is the maximal zero-dimensional subsheaf, and n_j is the length of its local Hilbert scheme. A generic geometric deformation has exactly n_j disjoint $(-1, -1)$ -curves in class $j[C]$.

- (v) *The contraction algebra satisfies*

$$\dim_{\mathbb{C}} A_{\text{con}} = \sum_{j=1}^{\ell} j^2 n_j.$$

Proof. The description of the bases and the marked curve divisors is the Katz–Morrison and Bryan–Katz–Leung simultaneous-resolution construction [5, Propositions 2.2–2.3 and Lemma 2.4]. For the formulation using restricted roots in general flopping contractions, see [17, Theorem 2.6]. Their Lemma 2.8 identifies the number of transverse intersections with D_j with the multiplicity k_j of the isolated Hilbert point $[C_j]$. Toda’s definition and deformation interpretation identify the integers n_j , and his noncommutative-width theorem gives the dimension formula [19, Section 2.4 and Theorem 1.1]. The equality of the two central thickenings is proved in Lemma 2.11 below. The relative-dimension bound, the identity $R\pi_{\ell*} \mathcal{O} = \mathcal{O}$, and finiteness of the universal algebra over its centre follow from the universal-flop construction [13, Section 2.3, Assumption 3.1, and Section 3.3]. $\square$

2.1 Restricted-root geometry and marked morsification

Let $q : \mathfrak{h} \rightarrow H$ be the quotient map. For a root α , let $H_\alpha \subset \mathfrak{h}$ be its root hyperplane. The group W_0 preserves the marked coefficient and permutes the non-Levi roots of each coefficient. If $\mathcal{O}$ is such an orbit, the square-free product

$$p_{\mathcal{O}} = \prod_{\alpha \in \mathcal{O}} \alpha$$

is W_0 -invariant and descends to an equation on H . The roots of marked coefficient j form a single W_0 -orbit $\mathcal{O}_j$ (Lemma 2.5), so $p_{\mathcal{O}_j}$ descends to a polynomial $P_j \in \mathcal{O}(H)$, with $q^*P_j = p_{\mathcal{O}_j}$, cutting out the reduced irreducible divisor D_j of Proposition 2.4(ii). Put

$$D^{\text{curv}} = \bigcup_{j=1}^{\ell} D_j;$$

this is the curve discriminant, the locus where the marked partial resolution has a complete exceptional curve.

Lemma 2.5 (The coefficient strata are single W_0 -orbits). *For each of the six primitive marked ADE pairs and each $1 \leq j \leq \ell$, the set*

$$\mathcal{O}_j := \{\alpha \in \Phi^+ : c_*(\alpha) = j\}$$

is a single W_0 -orbit. Its cardinalities are:

length ℓ	marked pair	$ \mathcal{O}_1 , \dots, \mathcal{O}_\ell $
1	A_1	1
2	D_4	8, 1
3	E_6	18, 9, 2
4	E_7	24, 18, 8, 3
5	$E_8, c_*(\theta) = 5$	40, 30, 20, 10, 4
6	$E_8, c_*(\theta) = 6$	30, 30, 20, 15, 6, 5

In particular $\sum_j |\mathcal{O}_j|$ is the number of positive roots with $c_(\alpha) > 0$, written N_p in Section 4.4, and each D_j is irreducible.*

Proof. The group W_0 preserves c_* , so it acts on each $\mathcal{O}_j$. The finite reflection calculation in Appendix A enumerates the roots and the W_0 -orbits, giving the displayed cardinalities and transitivity. Since W_0 acts transitively on $\mathcal{O}_j$, the descended equation $p_{\mathcal{O}_j}$ is irreducible on H : a proper W_0 -stable factor would correspond to a proper W_0 -stable subset of $\mathcal{O}_j$. Hence D_j is irreducible, and by the first assertion of Lemma 2.6 it is reduced. $\square$

Lemma 2.6 (Reduced root divisors and the higher-root locus). *Let α be a non-Levi root. The quotient $\mathfrak{h} \rightarrow H = \mathfrak{h}/W_0$ is étale at a general point of H_α . Consequently the divisor descended from every W_0 -orbit of non-Levi root hyperplanes is generically reduced.*

There is a closed subset $\Sigma^{\text{bad}} \subset D^{\text{curv}}$ of codimension at least two in H , containing:

- (i) *the singular loci of all curve-discriminant components;*

(ii) *intersections of distinct curve-discriminant components;*

(iii) *the images of intersections $H_\alpha \cap H_\beta$, where α is non-Levi and $\beta \neq \pm\alpha$.*

We call the points of $D^{\text{curv}} \setminus \Sigma^{\text{bad}}$ *good*. At each Cartan lift of a good point of D_j , exactly one root pair $\{\pm\alpha\}$ vanishes.

Proof. Choose

$$x \in H_\alpha \setminus \bigcup_{\beta \neq \pm\alpha} H_\beta.$$

Its stabilizer in W is $\{1, s_\alpha\}$. Since α is not a Levi root, $s_\alpha \notin W_0$, and the W_0 -stabilizer of x is trivial. A finite quotient in characteristic zero is étale at a point with trivial stabilizer. The product of the distinct linear root equations therefore descends with generic multiplicity one.

For $\beta \neq \pm\alpha$, the hyperplanes H_α and H_β are distinct, so their intersection has codimension two. A finite morphism preserves the dimension of an irreducible closed subset. Thus all images in (iii) have codimension at least two in H . Distinct reduced curve divisors have no common codimension-one component, and the singular locus of a reduced divisor in the smooth space H has ambient codimension at least two. Taking the finite union proves the claim. Outside this set, each Cartan lift lies on precisely one root hyperplane. For A_1 , there are no intersections of distinct root hyperplanes and the bad locus is empty. $\square$

Lemma 2.7 (Quadratic reflection coordinate). *Let $\bar{x} \in D_j \setminus \Sigma^{\text{bad}}$, and choose a Cartan lift $x \in H_\alpha$. There are completed local coordinates $(\zeta, z_2, \dots, z_r)$ at $\bar{x}$ and $(t, z_2, \dots, z_r)$ at its image in $\mathfrak{h}/W$ such that*

$$\widehat{\mathcal{O}}_{H, \bar{x}} = \mathbb{C}[[\zeta, z_2, \dots, z_r]], \quad \widehat{\mathcal{O}}_{\mathfrak{h}/W, \pi_W(\bar{x})} = \mathbb{C}[[t, z_2, \dots, z_r]],$$

and $\pi_W^* t = \zeta^2$, while $D_j = (\zeta = 0)$.

Proof. The W_0 -stabilizer is trivial, while the full Weyl stabilizer is the order-two group generated by s_α . This reflection acts by $\zeta \mapsto -\zeta$ on the normal line to H_α and trivially along the hyperplane. The invariant completed local ring is therefore $\mathbb{C}[[\zeta^2, z_2, \dots, z_r]]$. $\square$

Lemma 2.8 (Specialization of the marked degree). *Let $\mathcal{L}$ be a line bundle on the marked partial-resolution family, and let E_b be its exceptional curve over a good point $b \in D_j$. Then*

$$\deg_{E_b} \mathcal{L}_b = j \deg_C \mathcal{L}_0.$$

Proof. At the generic point of C , the normal central surface has a DVR as local ring. The symbolic j -thickening has generic length j , hence fundamental cycle $[C_j] = j[C]$. It follows that $\deg_{C_j} \mathcal{L}_0 = j \deg_C \mathcal{L}_0$. In the projective flat curve family $\mathcal{C}_j \rightarrow D_j$ of Proposition 2.4, both $\mathcal{O}_{\mathcal{C}_j}$ and the restriction of $\mathcal{L}$ are flat over the base. Their fibrewise Euler characteristics are locally constant. Their difference, which is the degree of $\mathcal{L}$ on a fibre, is therefore constant on the connected base germ. This gives the equality; compare [19, Section 2.4, equation (15)]. $\square$

Lemma 2.9 (The transverse threefold is an ordinary double point). *At a point of $D_j \setminus \Sigma^{\text{bad}}$, the transverse surface deformation is*

$$xy - z^2 = t.$$

After pullback to the marked partial-resolution base it is

$$xy - z^2 = \zeta^2,$$

and hence, after $\xi_+ = z + i\zeta$, $\xi_- = z - i\zeta$, it is the ordinary double point

$$xy - \xi_+\xi_- = 0.$$

If a one-dimensional classifying curve meets D_j transversely, its partial resolution contains one $(-1, -1)$ -curve in numerical class $j[C]$.

Proof. By the choice of Σ^{bad} , only the root subsystem $\{\pm\alpha\} \cong A_1$ vanishes. The simultaneous-resolution description is therefore, transversely to the root hyperplane, the semiuniversal A_1 -deformation [5, Proposition 2.2]. For completeness, the A_1 singularity is $xy - z^2 = 0$, its one-dimensional T^1 is generated by the constant class, and hence a miniversal equation is

$$xy - z^2 = t.$$

Lemma 2.7 gives $t = \zeta^2$. Since $(z + i\zeta)(z - i\zeta) = z^2 + \zeta^2$, the pulled-back equation is exactly $xy - \xi_+\xi_- = 0$. On a transverse classifying curve, ζ is a local parameter. Using it as the curve coordinate absorbs the higher terms in the original parametrization and gives the same completed ODP equation. The partial resolution is one of the two small resolutions of the ODP; its exceptional curve has normal bundle $\mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$. The degree comparison in Lemma 2.8 identifies its class as $j[C]$. $\square$

Lemma 2.10 (The central arc is not contained in the curve discriminant). *The image of f_0 is contained in no D_j . After shrinking to a closed disk $\overline{\mathbb{D}}_\rho$,*

$$f_0^{-1}(D^{\text{curv}}) \cap \overline{\mathbb{D}}_\rho = \{0\}, \quad f_0(\partial\mathbb{D}_\rho) \cap D^{\text{curv}} = \emptyset.$$

If P_j is a reduced equation of D_j , then

$$0 < m_j := \text{ord}_{w=0} P_j(f_0(w)) < \infty.$$

Proof. If the arc were contained in D_j , the pullback of the projective flat family $\mathcal{C}_j \rightarrow D_j$ would be a two-dimensional closed subset of the exceptional locus of the threefold contraction, contradicting $\text{Exc}(f) = C$. Thus each reduced local equation P_j pulls back to a nonzero holomorphic function whose zero at the origin is isolated. There are only finitely many j , so a common shrinking gives the last assertions. $\square$

Lemma 2.11 (Symbolic and pure thickenings coincide). *The Bryan–Katz–Leung symbolic thickening C_j equals Toda’s pure one-dimensional thickening $C^{(j)}$. Consequently their Hilbert multiplicities satisfy $k_j = n_j$.*

Proof. Work in a stalk $\mathcal{S}$ of the normal surface S_0 and let $\mathfrak{p} \subset \mathcal{S}$ be the height-one prime of C . By definition,

$$\mathfrak{p}^{(j)} = \mathfrak{p}^j \mathcal{S}_{\mathfrak{p}} \cap \mathcal{S}.$$

This is $\mathfrak{p}$ -primary, and $\mathfrak{p}^{(j)}/\mathfrak{p}^j$ is supported at closed points, hence has finite length. Since $\mathcal{S}/\mathfrak{p}^{(j)}$ has only the associated prime $\mathfrak{p}$, it has no nonzero finite-length submodule. In the exact sequence

$$0 \longrightarrow \mathfrak{p}^{(j)}/\mathfrak{p}^j \longrightarrow \mathcal{S}/\mathfrak{p}^j \longrightarrow \mathcal{S}/\mathfrak{p}^{(j)} \longrightarrow 0,$$

the first term is therefore the maximal finite-length submodule of $\mathcal{S}/\mathfrak{p}^j$, precisely Toda's Q . Thus $(\mathcal{O}/I^j)/Q = \mathcal{O}/I^{(j)}$, and the local Hilbert rings are identical. Proposition 2.4(iii)–(iv) gives $k_j = n_j$. $\square$

Lemma 2.12 (Parametric transversality with boundary). *Let M be a compact smooth manifold with boundary, let $\mathcal{U}$ and N be smooth manifolds, and let $S_1, \dots, S_m \subset N$ be smooth locally closed submanifolds. If a smooth map $\Xi: M \times \mathcal{U} \rightarrow N$ and its boundary restriction are transverse to every S_i , then the parameters for which both slice maps are simultaneously transverse to all S_i form a residual dense subset of $\mathcal{U}$. If $\dim_{\mathbb{R}} M = 2$, such a slice misses every stratum of real codimension at least four, and its boundary misses every stratum of real codimension at least two.*

Proof. For one stratum put $Z = \Xi^{-1}(S)$. Transversality makes Z a smooth incidence manifold, with boundary incidence Z_{∂} . A parameter is a regular value of $Z \rightarrow \mathcal{U}$ exactly when its slice is transverse to S ; the analogous statement holds on the boundary. Sard's theorem, applied on a countable exhaustion by relatively compact coordinate pieces, shows that the common regular values are residual and dense. Intersect these sets for the finitely many strata. The dimension formula for transverse inverse images gives the final avoidance statements. $\square$

Lemma 2.13 (A boundary-controlled constant translate). *There is an arbitrarily small vector $v \in H \cong \mathbb{C}^{\dim H}$ such that $f_v(w) = f_0(w) + v$ is transverse to every smooth codimension-one curve-discriminant stratum, avoids Σ^{bad} , and has no curve-discriminant point on $\partial\mathbb{D}_{\rho}$. The vector can be chosen so that, for some $\epsilon > 0$,*

$$f_0(w) + sv \notin D^{\text{curv}} \quad \text{for } w \in \partial\mathbb{D}_{\rho}, \quad |s| \leq 1 + \epsilon.$$

Proof. Choose a finite Whitney stratification of the curve discriminant adapted to Σ^{bad} . The translation spray

$$\overline{\mathbb{D}}_{\rho} \times H \longrightarrow H, \quad (w, \xi) \longmapsto f_0(w) + \xi$$

and its boundary restriction are submersions. Apply Lemma 2.12. A two-real-dimensional disk avoids the codimension-at-least-four bad strata, and its one-dimensional boundary avoids every curve-discriminant stratum. Since the compact boundary of the central disk is disjoint from the closed curve discriminant, reduced local equations have a positive minimum there. Uniform continuity supplies a ball of translations preserving boundary nonvanishing; choose the transverse parameter inside a slightly smaller ball. $\square$

Lemma 2.14 (Conservation of the root intersections). *Put $F(s, w) = f_0(w) + sv$ and $G_j(s, w) = P_j(F(s, w))$. On a connected s -disk U containing 0 and 1, the zero scheme of $G_j(s, \cdot)$ is finite flat of degree m_j and is represented by a monic polynomial*

$$W_j(s, T) \in \mathcal{O}(U)[T], \quad \deg_T W_j = m_j, \quad W_j(0, T) = T^{m_j}.$$

Proof. Boundary nonvanishing makes the argument-principle integral

$$\frac{1}{2\pi i} \int_{\partial\mathbb{D}_{\rho}} \frac{\partial_w G_j(s, w)}{G_j(s, w)} dw$$

constant in s , with value m_j . For $r \geq 1$, the analogous integral with an extra factor w^r is holomorphic in s and equals the r -th power sum of the zeros, counted with multiplicity.

Newton's identities therefore give holomorphic elementary symmetric functions on the whole parameter disk. Let $W_j(s, T)$ be the monic polynomial having these coefficients.

For each s , its roots and their multiplicities are exactly the zeros of $G_j(s, \cdot)$ in the w -disk. Local Weierstrass preparation at each zero implies that G_j/W_j extends holomorphically across the zero locus and is a unit. Thus the two equations define the same analytic scheme. The quotient by the monic polynomial is locally free with basis $1, T, \dots, T^{m_j-1}$, and boundary nonvanishing makes its support proper over the parameter disk. At $s = 0$, all m_j zeros lie at the origin, so $W_j(0, T) = T^{m_j}$. $\square$

Lemma 2.15 (Passage from one good fibre to the formal generic fibre). *Put $B = \mathbb{C}[[s, w]]$ and $K = \mathbb{C}((s))$. The polynomials $W_j \in K[T]$ are separable and pairwise coprime. The classifying map on $\text{Spec } B[1/s]$ avoids Σ^{bad} , and every root is transverse to the smooth part of its unique divisor D_j .*

Proof. At $s = 1$, the discriminant of each W_j and every pairwise resultant are nonzero. They are therefore nonzero holomorphic functions of s . Their Taylor series may be divisible by s , but become units after passing to K . This proves separability and coprimality.

For an irreducible component $S \subset \Sigma^{\text{bad}}$, let $I_S = F^{-1}(S)$. Since $S \subset D^{\text{curv}}$, this is a closed subspace of the finite root incidence $\bigcup_j \{G_j = 0\}$ from Lemma 2.14. Its image in U is a proper analytic subset, because its fibre at $s = 1$ is empty. The completed direct image of $\mathcal{O}_{I_S}$ at $s = 0$ therefore has finite length over $\mathbb{C}[[s]]$ and vanishes after inverting s . There are finitely many such components. Finally, at a smooth point of D_j , separability is exactly the condition $\partial_w G_j \neq 0$, hence transversality. $\square$

Proposition 2.16 (The conserved multiplicities are the GV invariants). *For each j ,*

$$m_j = n_j. \quad (2.2)$$

Thus the formal constant-translation family has exactly n_j coefficient- j roots.

Proof. Write $\tilde{f}$ for the boundary-proper generic perturbation of f_0 used by Bryan–Katz–Leung. For that perturbation, conservation of the Cartier intersection number gives $m_j = \#\tilde{f}^{-1}(D_j) = k_j$. Lemma 2.11 gives $k_j = n_j$. Lemma 2.14 gives the same intersection number for the constant-translation family. $\square$

Proposition 2.17 (Marked formal morsification). *There is a finite extension K'/K with valuation ring $\mathbb{C}[[u]]$, where $K' = \mathbb{C}((u))$ and $s = u^e$ for some $e \geq 1$. The completed local models at the generic curve points are small resolutions of split ordinary double points. Moreover,*

$$W_j(u^e, w) = \prod_{a=1}^{n_j} (w - w_{j,a}(u)), \quad w_{j,a}(u) \in u\mathbb{C}[[u]],$$

with all roots pairwise distinct, including roots belonging to different coefficients. The pullback of the universal partial resolution has exactly n_j pairwise disjoint $(-1, -1)$ -curves in class $j[C]$, and no other complete curves.

Proof. First split the finitely many separable polynomials W_j . The resulting curve points are geometrically ordinary double points by Lemma 2.9. Enlarge the field by a finite extension splitting their quadratic tangent cones. The formal Morse lemma then gives the split ODP

equations, and the small resolutions correspond to their two rulings. Denote the resulting field by K' . The completed normalization is a complete equicharacteristic DVR with residue field $\mathbb{C}$, hence is $\mathbb{C}[[u]]$; after a unit adjustment of u , its ramified parameter is $s = u^e$. Every root is integral because W_j is monic, and $W_j(0, T) = T^{n_j}$ forces its residue to be zero, so $w_{j,a}(u) \in u\mathbb{C}[[u]]$. Separability and coprimality give all the asserted distinctness. Lemma 2.9 supplies one $(-1, -1)$ -curve of class $j[C]$ at each root; different roots lie in different w -fibres and are disjoint.

To prove exhaustion, observe that the completed target is affine. A complete curve in the source maps to a point and hence lies in the exceptional locus; the regular function w is constant on it. The curve discriminant D^{curv} is the full locus of complete exceptional curves. Every generic point of the formal disk on that locus is a good point of one unique D_j , where the unique curve was already counted, and the disk avoids the codimension-two bad locus. Hence the list is exhaustive. $\square$

2.2 Universal tilting bundles and the coefficient- j block

Karmazyn's universal length- ℓ partial resolution carries the basic zero-perverse generator and its positive dual

$$\mathcal{T} = \mathcal{O} \oplus \mathcal{T}', \quad \mathcal{P} = \mathcal{T}^\vee = \mathcal{O} \oplus \mathcal{P}'.$$

Set

$$\mathcal{A}_\ell = \text{End}_{\mathcal{Y}_\ell}(\mathcal{T})^{\text{op}} \cong \text{End}_{\mathcal{Y}_\ell}(\mathcal{P}), \quad e_0 = [\mathcal{O}].$$

The universal source, target, centre, and algebra are flat over the marked base $H = \text{Spec } \mathbb{H}_\ell$ by the universal-flop construction [13, Sections 2.3 and 3.3]. The universal target is affine of finite type over H . The positive nonfree summand is the normalized maximal extension

$$0 \longrightarrow \mathcal{O}^{\oplus(\ell-1)} \longrightarrow \mathcal{P}' \longrightarrow \mathcal{L} \longrightarrow 0, \quad \deg_C \mathcal{L} = 1. \quad (2.3)$$

Thus $\text{rk } \mathcal{P}' = \ell$ and $\deg_C \det \mathcal{P}' = 1$. $\mathcal{P}$ belongs to the (-1) -perverse heart, while $\mathcal{T}$ belongs to the zero-perverse heart. Duality fixes the distinguished free projector and turns the endomorphism algebra into its opposite.

Lemma 2.18 (Tor-independence for arbitrary classifying pullback). *Let B be a Noetherian $\mathbb{H}_\ell$ -algebra, and form*

$$\begin{array}{ccc} \mathcal{Y}_B & \xrightarrow{g} & \mathcal{Y}_\ell \\ \pi_B \downarrow & & \downarrow \pi \\ \mathcal{X}_B & \xrightarrow{h} & \mathcal{X}_\ell. \end{array}$$

This square is Tor-independent. Consequently

$$R\pi_{B*} Lg^* E \simeq Lh^* R\pi_* E \quad (2.4)$$

for every perfect E on $\mathcal{Y}_\ell$, and $R\pi_{B} \mathcal{O}_{\mathcal{Y}_B} = \mathcal{O}_{\mathcal{X}_B}$.*

Proof. Locally write the rings as $C \rightarrow A \rightarrow Q$, where A and Q are flat over $C = \mathcal{O}_H$. Then

$$Q \otimes_A^L (A \otimes_C B) \simeq Q \otimes_C^L B \simeq Q \otimes_C B.$$

Thus the higher Tor groups vanish, and the standard derived base-change theorem gives (2.4). $\square$

Lemma 2.19 (Base change of the perverse generator). *The pullbacks $\mathcal{P}_B = g^*\mathcal{P}$ and $\mathcal{T}_B = g^*\mathcal{T}$ are projective generators of the (-1) - and zero-perverse hearts of π_B , respectively, and induce the corresponding equivalences on bounded coherent derived categories.*

Proof. Van den Bergh's category $\mathfrak{V}$ consists of globally generated bundles M satisfying $R^1\pi_*(M^\vee) = 0$. A finite evaluation surjection for $\mathcal{P}$ remains surjective after pullback. Since $\mathcal{P} \in \mathfrak{V}$, $R\pi_*\mathcal{T} = (\pi_*\mathcal{T})[0]$. By Lemma 2.18,

$$R\pi_{B*}\mathcal{T}_B \simeq Lh^*(\pi_*\mathcal{T}).$$

The left side lies in $D^{\geq 0}$, while the right side lies in $D^{\leq 0}$; hence both are concentrated in degree zero and $R^1\pi_{B*}\mathcal{T}_B = 0$. Relative ampleness of $\det \mathcal{P}$, global generation, and the direct $\mathcal{O}$ -summand also survive. Van den Bergh's intrinsic projective-generator criterion now applies [20, Propositions 3.2.5–3.2.7], and the derived equivalences follow from [20, Corollary 3.2.8]. $\square$

Proposition 2.20 (Ordinary endomorphism base change). *There is a canonical isomorphism of marked algebras*

$$\mathrm{End}_{\mathcal{Y}_B}(\mathcal{T}_B)^{\mathrm{op}} \cong \mathcal{A}_\ell \otimes_{\mathbb{H}_\ell} B. \quad (2.5)$$

All higher self-Ext groups vanish. The isomorphism preserves e_0 , and the bundle pullbacks preserve ranks and determinants. The algebra identification remains valid after central localization and Noetherian completion.

Proof. Apply derived base change to the vector bundle $\mathcal{E}nd(\mathcal{T})$. Tilting gives

$$R\pi_*\mathcal{E}nd(\mathcal{T}) = \pi_*\mathcal{E}nd(\mathcal{T})[0].$$

Its module of sections is the $\mathbb{H}_\ell$ -flat module underlying $\mathcal{A}_\ell^{\mathrm{op}}$. Hence its derived pullback is its ordinary pullback in degree zero. Taking sections proves (2.5) and the Ext vanishing. The distinguished free summand is pulled back, and rank and determinant commute with pullback. Localization and Noetherian completion are flat, so the same argument applies after either operation. $\square$

Write $\Lambda_B = \mathcal{A}_\ell \otimes_{\mathbb{H}_\ell} B$. Its quotient by $\Lambda_B e_0 \Lambda_B$ is the relative contraction quotient.

Proposition 2.21 (The special fibre is the basic NCCR). *Let $B = \mathbb{C}[[s, w]]$ carry a classifying map whose restriction modulo s is the central arc. Localize Λ_B at the central contraction point of its centre and complete there. The reduction $\widehat{\Lambda_B}/(s)$ is isomorphic to the basic NCCR of the original flop, preserving the free idempotent e_0 .*

Proof. The algebra is finite over its Noetherian centre, so completion commutes with reduction modulo s . Lemma 2.3 identifies the resulting completed geometry with the original flop.

On the completed one-curve flop, Van den Bergh's classification gives exactly two indecomposable positive perverse projectives: $\mathcal{O}$ and the maximal extension $\mathcal{M}$, with $\mathrm{rk} \mathcal{M} = \ell$ and $\deg_C \det \mathcal{M} = 1$ [20, Proposition 3.5.4 and Theorem 3.5.5]. Write the reduced nonfree universal summand as $\mathcal{O}^{\oplus a} \oplus \mathcal{M}^{\oplus b}$. Determinant degree gives $b = 1$, while rank gives $\ell = a + \ell$, hence $a = 0$. Thus the reduced positive generator is $\mathcal{O} \oplus \mathcal{M}$; dualizing gives the selected zero-perverse basic generator and fixes e_0 . $\square$

We now work at a geometric generic coefficient- j point after the finite extension of Proposition 2.17. Denote the point by λ , its completed small resolution by $\hat{Y}_\lambda \rightarrow \text{Spec } \hat{R}_\lambda$, and its exceptional curve by C_λ . The local model is split over K' .

Lemma 2.22 (The split ODP projectives over the splitting field). *Let F be any characteristic-zero field and put*

$$R_F = F[[a_1, a_2, b_1, b_2]]/(a_1a_2 - b_1b_2), \quad I = (a_1, b_1).$$

On the small resolution $\pi_Z: Z = \text{Proj}_{R_F} \bigoplus_{n \geq 0} I^n \rightarrow \text{Spec } R_F$, let $\mathcal{Q} = \mathcal{O}_Z(1)$. Then $\mathcal{O}_Z \oplus \mathcal{Q}$ is the basic positive perverse projective generator. Every positive perverse projective vector bundle is a direct sum of $\mathcal{O}_Z$ and $\mathcal{Q}$, and

$$\text{rk } \mathcal{Q} = 1, \quad \deg_C \mathcal{Q} = 1, \quad \pi_{Z*} \mathcal{Q} = I.$$

These assertions remain true after completion along the exceptional curve.

Proof. Write $\theta = b_1/a_1$ on the a_1 -chart and $\theta^{-1} = a_1/b_1$ on the b_1 -chart. After removing the Rees torsion and completing along the exceptional curve, the charts and their gluing are

$$\begin{aligned} \hat{U}_{a_1} &= \text{Spf } F[\theta][[a_1, b_2]], \quad b_1 = a_1\theta, \quad a_2 = \theta b_2, \\ \hat{U}_{b_1} &= \text{Spf } F[\theta^{-1}][[b_1, a_2]], \quad a_1 = \theta^{-1}b_1, \quad b_2 = \theta^{-1}a_2. \end{aligned}$$

Thus Z is the split Atiyah small resolution. Its exceptional curve is given by $a_1 = b_2 = 0$ on $\hat{U}_{a_1}$ and $b_1 = a_2 = 0$ on $\hat{U}_{b_1}$, so $C \cong \mathbb{P}_F^1$. The transition of the tautological quotient is the transition of $\mathcal{O}_{\mathbb{P}^1}(1)$; consequently $\mathcal{Q}|_C = \mathcal{O}_C(1)$. The quotient $R_F/I \cong F[[a_2, b_2]]$ and the depth lemma show that I is maximal Cohen–Macaulay, hence reflexive. The pushforward of $\mathcal{Q}$ is reflexive by Hartogs extension across C in the smooth small resolution. It agrees with I away from the vertex, so $\pi_{Z*} \mathcal{Q} = I$.

Here $N_{C/Z} \cong \mathcal{O}_C(-1)^{\oplus 2}$. On the n -th infinitesimal graded piece of the formal neighbourhood of C , the restrictions of $\mathcal{O}_Z$ and $\mathcal{Q}^{-1}$ are respectively direct sums of

$$\text{Sym}^n N_{C/Z}^\vee \cong \mathcal{O}_C(n)^{\oplus(n+1)} \quad \text{and} \quad \mathcal{O}_C(n-1)^{\oplus(n+1)}.$$

Their first cohomology vanishes for every $n \geq 0$. Induction over the infinitesimal thickenings, followed by the theorem on formal functions, therefore gives

$$R^1 \pi_{Z*} \mathcal{O}_Z = R^1 \pi_{Z*} \mathcal{Q}^{-1} = 0.$$

The algebraic small resolution of the affine quadratic cone is the total space of $\mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$; the charts above describe its completion along the zero section. The bundle $\mathcal{Q}$ is relatively ample and globally generated. Together with the cohomology vanishings and the structure-sheaf summand, these properties verify Van den Bergh’s projective-generator criterion [20, Propositions 3.2.5–3.2.7], both over the complete local target and on the formal completion.

Its endomorphism algebra is finite over the complete local ring R_F and is semiperfect. The two summands have local endomorphism rings $\text{End}(\mathcal{O}_Z) = R_F = \text{End}(\mathcal{Q})$, and their degrees on C distinguish them. Their projectors therefore form a complete set of pairwise inequivalent primitive idempotents. Projectives over a semiperfect ring are direct sums of the corresponding indecomposable projectives. The tilting equivalence therefore proves the decomposition of every projective in the heart. The two summands and their idempotents are defined over F , and the same argument applies to the formal completion. $\square$

Proposition 2.23 (Decomposition of the local projective generator). *Let $\mathcal{Q}$ be the unique nonfree indecomposable positive perverse projective on the completed ODP small resolution, normalized by $\mathrm{rk} \mathcal{Q} = 1$ and $\deg \det \mathcal{Q}|_{C_\lambda} = 1$. Then*

$$\widehat{\mathcal{P}}'_\lambda \cong \mathcal{O}^{\oplus(\ell-j)} \oplus \mathcal{Q}^{\oplus j}. \quad (2.6)$$

After pushforward to the ODP centre,

$$N_\lambda \cong \widehat{R}_\lambda^{\oplus(\ell-j)} \oplus I_\lambda^{\oplus j},$$

where I_λ is the rank-one Atiyah MCM module.

Proof. Proposition 2.20 and Lemma 2.19 apply after field specialization and completion. The roots have been split over K' , so the completed local model is the split ODP of Lemma 2.22 with $F = K'$. That lemma, followed by Krull–Schmidt, gives $\widehat{\mathcal{P}}'_\lambda \cong \mathcal{O}^{\oplus a} \oplus \mathcal{Q}^{\oplus b}$. Rank gives $a + b = \ell$. Determinant commutes with pullback, and Lemma 2.8 gives

$$\deg_{C_\lambda} \det \widehat{\mathcal{P}}'_\lambda = j \deg_C \det \mathcal{P}' = j.$$

The same degree on the decomposition is b . Thus $b = j$ and $a = \ell - j$. Lemma 2.22 identifies the pushforward of $\mathcal{Q}$ with the Atiyah module I_λ . $\square$

For a module over a local ring R , write $\underline{\mathrm{End}}_R$ for endomorphisms modulo maps factoring through finite free R -modules.

Lemma 2.24 (Stable endomorphisms of the Atiyah module). *Let*

$$R_{\mathrm{odp}} = F[[a_1, a_2, b_1, b_2]]/(a_1 a_2 - b_1 b_2), \quad I = (a_1, b_1),$$

where F has characteristic zero. Then

$$\underline{\mathrm{End}}_{R_{\mathrm{odp}}}(I) \cong F, \quad \underline{\mathrm{End}}_{R_{\mathrm{odp}}}(I^{\oplus j}) \cong M_j(F).$$

Proof. Write $R = R_{\mathrm{odp}}$. Since I is rank-one reflexive over the normal domain R , its endomorphisms are multiplications by fractions. At each height-one prime, I is free of rank one; hence $(I : I) \subseteq \bigcap_{\mathrm{ht} p=1} R_p = R$. Thus $\underline{\mathrm{End}}_R(I) = R$.

We claim that

$$I^{-1} = R + R(b_2/a_1).$$

The displayed right-hand side is contained in I^{-1} , since $(b_2/a_1)a_1 = b_2$ and $(b_2/a_1)b_1 = a_2$. Conversely, if $qI \subset R$, put $r = qa_1 \in R$. Then $b_1 r \in a_1 R$. In $R/(a_1) = F[[a_2, b_1, b_2]]/(b_1 b_2)$, the annihilator of b_1 is (b_2) . Thus $r = a_1 r_0 + b_2 r_1$, and $q = r_0 + (b_2/a_1)r_1$, proving the claim. Endomorphisms factoring through a free module form the ideal

$$II^{-1} = (a_1, b_1) + (b_2, a_2) = (a_1, a_2, b_1, b_2) = \mathfrak{m}.$$

Therefore $\underline{\mathrm{End}}_R(I) = R/\mathfrak{m} = F$, and matrices give the second assertion. $\square$

Proposition 2.25 (Idempotent-preserving resolved/MCM comparison). *For the completed ODP small resolution, pushforward gives an isomorphism*

$$\mathrm{End}_{\widehat{Y}_\lambda}(\widehat{\mathcal{P}}_\lambda) \cong \mathrm{End}_{\widehat{R}_\lambda}(\widehat{R}_\lambda^{\oplus(\ell-j+1)} \oplus I_\lambda^{\oplus j}) \quad (2.7)$$

which preserves the projector onto the distinguished free summand.

Proof. The completed ODP is a normal complete local Gorenstein threefold, and the small resolution is smooth, projective, birational, and crepant, with fibres of dimension at most one and $R\hat{\pi}_*\mathcal{O} = \hat{R}$. The positive bundle is the perverse projective generator by Lemma 2.19; its two required $R^1\hat{\pi}_*$ -vanishings follow from the defining $\mathfrak{V}$ -condition and global generation. Van den Bergh's comparison theorem therefore identifies the resolved endomorphism algebra with that of the pushed-forward MCM generator [20, Lemma 3.2.9 and Proposition 3.2.10]. Naturality sends the direct-summand projector to the distinguished free projector. $\square$

Theorem 2.26 (The completed coefficient- j factor). *At every transverse coefficient- j point used in the marked formal morsification, the completion of the relative contraction quotient is the matrix algebra $M_j(K')$. The maximal ideal of the completed centre acts trivially on this factor.*

Proof. After the arbitrary classifying pullback, central localization and Noetherian completion are flat. For a finite algebra Λ with idempotent e_0 ,

$$\widehat{\Lambda/\Lambda e_0\Lambda} \cong \hat{\Lambda}/\hat{\Lambda}e_0\hat{\Lambda}.$$

By Proposition 2.25, this is the endomorphism algebra of $\hat{R}^{\ell-j+1} \oplus I^j$ modulo the ideal generated by one distinguished free projector. Every other free projector factors through that copy, so the quotient is the stable endomorphism algebra of I^j . Lemma 2.24 gives $M_j(K')$. The centre acts through the residue field, proving the last assertion. $\square$

2.3 Global support, finite freeness, and the existence theorem

Put $B = \mathbb{C}[[s, w]]$, and let $\nu : \text{Spec } B \rightarrow H$ be the completion of the constant-translation family $F(s, w) = f_0(w) + sv$ of Lemmas 2.13 and 2.14, and set

$$R_B = \mathcal{R}_\ell \otimes_{\mathbb{H}_\ell} B.$$

Let $\mathfrak{n} \subset R_B$ be the maximal ideal of the central contraction point (it contains s), and define the central completion

$$S_\nu = \widehat{(R_B)_\mathfrak{n}}.$$

Let $\mathcal{T}_\nu$ be the completed pullback of the tilting bundle. The corresponding algebra and contraction quotient are

$$\Lambda_\nu = \mathcal{A}_\ell \otimes_{\mathcal{R}_\ell} S_\nu \cong \text{End}(\mathcal{T}_\nu)^{\text{op}}, \quad \mathcal{A} = \Lambda_\nu / \Lambda_\nu e_0 \Lambda_\nu. \quad (2.8)$$

Proposition 2.27 (Identification of the special fibre). *There is a marked algebra isomorphism*

$$\mathcal{A}/(s) \cong A_{\text{con}}.$$

Proof. Proposition 2.21 gives $\Lambda_\nu/(s) \cong \Lambda$, the selected basic NCCR of the original flop, preserving e_0 . Right exactness gives

$$\mathcal{A}/(s) \cong \frac{\Lambda}{\Lambda e_0 \Lambda} = A_{\text{con}}.$$

$\square$

Lemma 2.28 (Fullness away from the root support). *Over the geometric generic marked slice, the relative contraction quotient vanishes away from the finitely many transverse curve-root points. Consequently it is a finite algebra supported on exactly those closed points.*

Proof. By Proposition 2.4(ii), D^{curv} is the full locus where a complete curve occurs. Away from D^{curv} , a positive-dimensional fibre of the projective morphism would contain a complete curve, so the morphism is quasi-finite and hence finite. Since $R\pi_{\ell*}\mathcal{O}_{\mathcal{Y}_{\ell}} = \mathcal{O}_{\mathcal{X}_{\ell}}$, a finite base-changed morphism is an isomorphism. Proposition 2.17 meets D^{curv} only at the good transverse roots and supplies exactly the curves already counted. Away from those points, every summand of the pushed-forward vector bundle is consequently locally free. After shrinking at a local point it is a finite-free module, so every matrix unit of its endomorphism algebra factors through the distinguished rank-one free summand. Hence e_0 is full and the idempotent quotient vanishes. At every transverse point, Theorem 2.26 supplies a nonzero matrix block. The universal endomorphism algebra is finite over its centre, and this property survives base change, localization, completion, and quotient. Thus the generic quotient is finite over a Noetherian centre with the stated finite support. $\square$

Lemma 2.29 (Assembly of the local factors). *Let S be a Noetherian K' -algebra and A a finite S -algebra supported at K' -rational maximal ideals $p_1, \dots, p_r$, so $S/p_i = K'$. Suppose*

$$\widehat{A_{p_i}} \cong M_{d_i}(K')$$

and the maximal ideal of $\widehat{S_{p_i}}$ acts trivially. Then

$$A \cong \prod_{i=1}^r M_{d_i}(K').$$

In particular its scheme-theoretic support is reduced.

Proof. Completion $S_{p_i} \rightarrow \widehat{S_{p_i}}$ is faithfully flat. Since $p_i \widehat{A_{p_i}} = 0$, faithful flatness gives $p_i A_{p_i} = 0$. Hence A_{p_i} already equals its completion and is $M_{d_i}(K')$. Put $J = \bigcap_i p_i$. The finite module JA localizes to zero away from the support and also at each p_i , where all other maximal ideals become units. Thus $JA = 0$. The Chinese remainder theorem gives $S/J \cong (K')^r$; its central idempotents split A , and the local factors are the stated matrix algebras. $\square$

Proposition 2.30 (The geometric generic fibre). *After the finite splitting extension $K'/\mathbb{C}((s))$ of Proposition 2.17,*

$$\mathcal{A} \otimes_{\mathbb{C}[[s]]} K' \cong \prod_{j=1}^{\ell} M_j(K')^{\times n_j}. \quad (2.9)$$

Proof. There are exactly n_j coefficient- j points by Propositions 2.16 and 2.17. In the normalized ramified model of that proposition, $K' = \mathbb{C}((u))$, $s = u^e$, and every root lies in $u\mathbb{C}[[u]]$. Thus the split points are K' -rational points of the formal w -disk, all specializing to its central maximal ideal. Lemma 2.28 gives the complete support, and Theorem 2.26 identifies every completed factor with $M_j(K')$. Apply Lemma 2.29 simultaneously to all support points. $\square$

Lemma 2.31 (Topological finiteness). *Let $R_0 = \mathbb{C}[[s]]$, let*

$$S = R_0[[x_1, \dots, x_m]]/J$$

be complete Noetherian, and let M be finite over S . If M/sM is finite-dimensional over $\mathbb{C}$, then M is finite over R_0 .

Proof. Choose lifts $m_1, \dots, m_d$ of a $\mathbb{C}$ -basis of M/sM . For $m \in M$, successive reduction modulo sM gives

$$m = \sum_{i=1}^d \left(\sum_{a=0}^{q-1} c_{ia} s^a \right) m_i + s^q m^{(q)}, \quad c_{ia} \in \mathbb{C}.$$

The series $c_i(s) = \sum_{a \geq 0} c_{ia} s^a$ belong to R_0 . For each q , their tails are divisible by s^q , so

$$m - \sum_i c_i(s) m_i \in s^q M.$$

The ring S is Noetherian local, s lies in its maximal ideal, and M is finite over S . Krull intersection gives $\bigcap_q s^q M = 0$. Hence the m_i generate M over R_0 . $\square$

Proposition 2.32 (Finiteness of the relative contraction algebra). *The algebra $\mathcal{A}$ is a finite $\mathbb{C}[[s]]$ -module.*

Proof. The universal target is of finite type over H , so its pullback is of finite type over $B = \mathbb{C}[[s, w]]$. Completion at the central contraction point therefore gives a ring of the form

$$S_\nu \cong \mathbb{C}[[s, x_1, \dots, x_m]]/J$$

for finitely many generators. Proposition 2.4(i) states that $\mathcal{A}_\ell$ is finite over $\mathcal{R}_\ell$. Base change and completion therefore make Λ_ν finite over S_ν , and its idempotent quotient $\mathcal{A}$ remains finite over S_ν .

By Proposition 2.27, $\mathcal{A}/(s) \cong A_{\text{con}}$ is finite-dimensional. Lemma 2.31 now gives finiteness over $\mathbb{C}[[s]]$. $\square$

Lemma 2.33 (DVR torsion exclusion). *Let M be a finite module over the DVR $\mathbb{C}[[s]]$. If*

$$\dim_{\mathbb{C}} M/sM = \dim_{\mathbb{C}((s))} M[1/s] = r,$$

then M is free of rank r .

Proof. The structure theorem gives

$$M \cong \mathbb{C}[[s]]^{\oplus r} \oplus \bigoplus_{h=1}^m \mathbb{C}[[s]]/(s^{e_h}).$$

The generic dimension is the free rank, while reduction modulo s has dimension $r + m$. Equality forces $m = 0$. $\square$

Proof of Theorem 2.1. The construction is (2.8). Proposition 2.27 gives its special fibre, and Proposition 2.30 gives its geometric generic product. In particular

$$\dim_{\mathbb{C}((s))} \mathcal{A}[1/s] = \sum_{j=1}^{\ell} j^2 n_j.$$

Proposition 2.32 gives finiteness over $\mathbb{C}[[s]]$. On the other hand, Proposition 2.4(v) and the special-fibre identification give

$$\dim_{\mathbb{C}} \mathcal{A}/(s) = \dim_{\mathbb{C}} A_{\text{con}} = \sum_{j=1}^{\ell} j^2 n_j.$$

Lemma 2.33 now proves that $\mathcal{A}$ is finite free of this rank. $\square$

3 An orbit criterion for semisimple deformations

We give a fixed-unit formulation of the orbit-dimension criterion used in [12, Section 1 and Theorem 2.14]. Let k be algebraically closed of characteristic zero, let $V = k \cdot 1 \oplus W$ have dimension d , and let X_d be the affine scheme of associative multiplications on V for which the distinguished vector 1 is the identity. The scheme X_d is defined by the associativity equations. The group preserving the identity is

$$G = \text{Stab}_{\text{GL}(V)}(1).$$

For a unital algebra Λ , the normalized Hochschild cochains are

$$C_{\text{norm}}^n(\Lambda, \Lambda) = \text{Hom}_k((\Lambda/k \cdot 1)^{\otimes n}, \Lambda).$$

We write Z_{norm}^n and B_{norm}^n for the cocycles and coboundaries of this complex.

Lemma 3.1 (Fixed-unit tangent spaces). *For $\Lambda \in X_d(k)$,*

$$T_\Lambda X_d = Z_{\text{norm}}^2(\Lambda, \Lambda), \quad T_\Lambda(G \cdot \Lambda) = B_{\text{norm}}^2(\Lambda, \Lambda),$$

and

$$\text{Lie Stab}_G(\Lambda) = \text{Der}(\Lambda).$$

Moreover, G is smooth, connected, and irreducible of dimension $d(d-1)$, and its orbits are precisely the unital isomorphism classes.

Proof. A dual-number multiplication $m + \varepsilon\phi$ retains the fixed unit exactly when ϕ is normalized; its linearized associativity equation is the normalized Hochschild cocycle equation. Differentiating transport by $1 + \varepsilon f$, with $f(1) = 0$, gives the normalized coboundaries, whose kernel consists of derivations.

Relative to $k \cdot 1 \oplus W$, every element of G has the unique form

$$\begin{pmatrix} 1 & \lambda \\ 0 & h \end{pmatrix}, \quad \lambda \in W^*, \quad h \in \text{GL}(W).$$

Thus $G \cong W^* \rtimes \text{GL}(W)$, proving the group assertions. A unital algebra isomorphism fixes the unique identity, so it belongs to G . $\square$

Lemma 3.2 (A family gives fixed-unit orbit closure). *Let $\mathfrak{o}$ be a complete equicharacteristic DVR with algebraically closed residue field k , a chosen coefficient field $k \subset \mathfrak{o}$, and fraction field K . Let $\mathcal{C}$ be a finite-free unital $\mathfrak{o}$ -algebra with special fibre A . If S is a d -dimensional k -algebra and*

$$\mathcal{C}_K \otimes_K \overline{K} \cong S \otimes_k \overline{K},$$

then

$$A \in \overline{G \cdot S} \subset X_d.$$

Proof. A uniformizer identifies $\mathfrak{o}$ with $k[[\varpi]]$. The unit of $\mathcal{C}$ is unimodular: its reduction is the nonzero unit of A . Extend it to an $\mathfrak{o}$ -basis lifting the prescribed basis of A . The structure constants then define a morphism $\text{Spec } \mathfrak{o} \rightarrow X_d$ whose closed point is A .

The geometric generic-fibre isomorphism is specified by finitely many entries algebraic over K , and therefore descends to a finite extension K'/K . The integral closure $\mathfrak{o}'$ of $\mathfrak{o}$ in K' is a finite complete DVR. Its residue field is a finite extension of the algebraically closed field k , hence is k . The generic-fibre isomorphism fixes the unit, so the base-changed generic point is a $G(K')$ -translate of S .

Let $Y = \overline{G \cdot S} \subset X_d$ with its reduced closed structure. Every equation of Y pulls back to zero in K' , hence to zero in the domain $\mathfrak{o}'$. The base-changed arc therefore factors through Y , and its closed point is A . This proves the orbit-closure relation. $\square$

Proposition 3.3 (Openness of semisimple orbits). *If $S \in X_d(k)$ is split semisimple, then S is a smooth point of the scheme X_d and $G \cdot S$ is open in X_d .*

Proof. The algebra S is separable, hence projective as an S^e -module. Thus $HH^i(S, S) = 0$ for $i > 0$, and in particular

$$Z_{\text{norm}}^2(S, S) = B_{\text{norm}}^2(S, S).$$

By Lemma 3.1, the tangent space of X_d at S equals the orbit tangent space.

The orbit is a smooth locally closed subvariety. Indeed, in characteristic zero its stabilizer is smooth, and the orbit is the homogeneous space $G/\text{Stab}_G(S)$; cf. [3, Proposition 1.11]. Near S , write $\mathcal{O} = \mathcal{O}_{X_d, S}$ and the orbit local ring as $\mathcal{O}/I$. If the common tangent/orbit dimension is δ , then $\dim(\mathcal{O}/I) = \delta$, while the embedding dimension of $\mathcal{O}$ is δ . Hence $\dim \mathcal{O} = \delta = \text{edim } \mathcal{O}$, so $\mathcal{O}$ is regular and therefore a domain. Since $\mathcal{O}/I$ has the same dimension and I is prime, $I = 0$. Thus the orbit contains an open neighbourhood of S ; translating that neighbourhood by G proves that the orbit is open. $\square$

Theorem 3.4 (Orbit-theoretic smoothness and uniqueness). *Let $A, S \in X_d(k)$, with S split semisimple. Assume*

$$A \in \overline{G \cdot S}, \quad \dim T_A X_d = \dim(G \cdot S). \quad (3.1)$$

Then:

- (i) A is a smooth point of X_d ;
- (ii) $\overline{G \cdot S}$ is the unique irreducible component of X_d through A ;
- (iii) if S' is split semisimple and $A \in \overline{G \cdot S'}$, then $S' \cong S$.

Proof. By Proposition 3.3 and the irreducibility of G , the orbit $G \cdot S$ is irreducible and open. Its closure is therefore an irreducible component $\mathcal{Z}$ of dimension $\delta = \dim(G \cdot S)$. Since $A \in \mathcal{Z}$, $\dim_A X_d \geq \delta$. The second hypothesis of (3.1) gives

$$\dim_A X_d \leq \dim T_A X_d = \delta.$$

Thus the local dimension equals the embedding dimension, so $\mathcal{O}_{X_d, A}$ is regular. Over a characteristic-zero field this is equivalent to smoothness at the k -point.

A regular local ring is a domain and has one minimal prime. Hence exactly one irreducible component of X_d passes through A , namely $\mathcal{Z}$. If S' is another semisimple generalization, the same openness argument makes $\overline{G \cdot S'}$ an irreducible component through A , so it also equals $\mathcal{Z}$. The two orbits are nonempty open subsets of the irreducible space $\mathcal{Z}$, hence intersect. Group orbits are equal or disjoint, so $G \cdot S' = G \cdot S$. Lemma 3.1 identifies this equality with a unital algebra isomorphism. $\square$

Corollary 3.5 (Hochschild numerical criterion). *Let $A, S \in X_d(k)$, with S semisimple and $A \in \overline{G \cdot S}$. The dimension equality in Theorem 3.4 is equivalent to*

$$\dim \operatorname{Der}(A) - \dim HH^2(A) = d - \dim Z(S). \quad (3.2)$$

In particular, it holds if

$$\dim Z(A) = \dim Z(S), \quad \dim HH^1(A) = \dim HH^2(A).$$

Proof. Lemma 3.1 gives

$$\dim T_A X_d = d(d-1) - \dim \operatorname{Der}(A) + \dim HH^2(A).$$

For split semisimple S , all derivations are inner and $\dim \operatorname{Der}(S) = d - \dim Z(S)$. Therefore

$$\dim(G \cdot S) = d(d-1) - (d - \dim Z(S)),$$

where characteristic zero ensures that the stabilizer group scheme is smooth, so its dimension equals the dimension of its Lie algebra. This proves the equivalence. Finally, $\dim \operatorname{Der}(A) = d - \dim Z(A) + \dim HH^1(A)$. $\square$

4 The cocenter and semisimple uniqueness

We compute the cocenter of the contraction algebra from the reduced curve discriminant. Symmetry and bimodule periodicity then turn its dimension into the tangent-space equality required by Section 3.

4.1 Statement and proof strategy

For the completed simple one-curve flop $f: X \rightarrow \operatorname{Spec} R$, let $A = A_{\text{con}}$ be its contraction algebra with $d = \dim_{\mathbb{C}} A$, and let n_j be its genus-zero GV invariants. Put

$$S_{\text{GV}} := \prod_{j=1}^{\ell} M_j(\mathbb{C})^{\times n_j}.$$

Theorem 4.1 (General semisimple–GV uniqueness). *For every completed simple one-curve flop of a smooth complex threefold,*

$$\dim_{\mathbb{C}} HH_0(A) = \dim_{\mathbb{C}} Z(A) = \sum_{j=1}^{\ell} n_j. \quad (4.1)$$

Consequently A is a smooth point of the fixed-unit associative-law scheme X_d , the orbit closure $\overline{G \cdot S_{\text{GV}}}$ is the unique irreducible component of X_d through A , and S_{GV} is the unique split semisimple complex deformation target of A . More generally, if $\mathcal{D}$ is a finite-free unital $\mathbb{C}[[u]]$ -algebra with $\mathcal{D}/u\mathcal{D} \cong A$ and geometrically semisimple generic fibre, then that generic fibre has geometric Wedderburn type S_{GV} .

The orbit criterion reduces uniqueness to a tangent-dimension equality. Symmetry and periodicity express this dimension in terms of $HH_0(A)$. We compute that cocenter by restricting the universal marked presentation to the classifying arc; its determinant records the intersections with the reduced curve divisors.

4.2 Structural reduction: symmetry, periodicity, Tate duality

Proposition 4.2 (Symmetry). *The contraction algebra is symmetric: $A_{\text{con}} \cong DA_{\text{con}}$ as A_{con} -bimodules.*

Proof. The completed base R is an isolated cDV singularity. If N is the nonfree Van den Bergh summand, then $A = \underline{\text{End}}_R(N)^{\text{op}}$. The stable endomorphism algebra is symmetric by [1, Theorem 2.6], and the opposite of a symmetric algebra is symmetric. $\square$

Proposition 4.3 (Bimodule four-periodicity). *In the stable category of A -bimodules,*

$$\Omega_{A^e}^4(A) \simeq A.$$

Proof. Let $\mathcal{B} = \text{CM}(R)$, whose projective-injective objects are add R . Write the Van den Bergh tilting bundle on the smooth source X as $\mathcal{O}_X \oplus \mathcal{N}$, and put $N = f_*\mathcal{N}$. By [11, Theorem 1.3.1], N is a $2\mathbb{Z}$ -cluster-tilting object of $\underline{\text{CM}}(R)$, with stable endomorphism algebra A^{op} . Thus $R \oplus N$ is a 2-cluster-tilting object of the Frobenius category $\mathcal{B}$.

The hypersurface identity $\Omega_R^2 \simeq \text{Id}$ gives the parameters $d_D = 2$ and $r_D = 1$ in Dugas's [7, Theorem 1.1(3)]. If $A \neq \mathbb{C}$, the one-curve contraction algebra is local and nonsemisimple, so it has no semisimple block. Dugas's theorem yields stable bimodule period dividing $(d_D + 2)r_D = 4$ for A^{op} , and hence for A , by taking opposite bimodules. If $A = \mathbb{C}$, its stable bimodule class is zero and its positive Hochschild groups vanish. $\square$

Lemma 4.4 (Tate–Hochschild duality). *For every symmetric finite-dimensional algebra A over $\mathbb{C}$,*

$$(\widehat{HH}^{-n}(A))^{\vee} \cong \widehat{HH}^{n-1}(A) \quad (n \in \mathbb{Z}).$$

If A is four-periodic in the stable bimodule category, then

$$DHH^1(A) \cong HH^2(A), \quad \dim HH^1(A) = \dim HH^2(A). \quad (4.2)$$

Proof. For a symmetric algebra A , the enveloping algebra A^e is symmetric, and $\widehat{HH}^*(A) = \widehat{\text{Ext}}_{A^e}^*(A, A)$. The displayed duality is [16, equation (2.11)]. Taking $n = 3$ and using four-periodicity gives

$$DHH^1(A) \cong D\widehat{HH}^{-3}(A) \cong \widehat{HH}^2(A) = HH^2(A),$$

because positive Tate degrees agree with ordinary Hochschild cohomology. $\square$

Lemma 4.5 (Centre–cocenter duality). *If A is finite-dimensional symmetric with symmetrizing trace $\tau: A \rightarrow \mathbb{C}$, then*

$$Z(A) \xrightarrow{\sim} HH_0(A)^{\vee}, \quad z \mapsto (\bar{a} \mapsto \tau(za)).$$

Consequently,

$$\dim Z(A) = \dim HH_0(A). \quad (4.3)$$

Proof. For $z \in Z(A)$,

$$\tau(z[a, b]) = \tau(zab - zba) = \tau(zab - azb) = 0,$$

so the functional factors through $A/[A, A]$. Conversely, identify $A \cong DA$ by the symmetrizing form. If the functional associated to $c \in A$ annihilates all commutators, then for all a, b ,

$$0 = \tau(c[a, b]) = \tau((ca - ac)b).$$

Nondegeneracy forces $ca = ac$ for all a , so $c \in Z(A)$. This proves perfection. $\square$

4.3 Fixed-unit tangent and orbit dimensions

Retain the fixed-unit scheme X_d and its group $G = \text{Stab}_{\text{GL}_d}(1)$ from Section 3, now over $\mathbb{C}$. In particular, $\dim G = d(d-1)$.

Lemma 4.6 (Tangent formula). *The Zariski tangent space is*

$$T_A X_d = Z_{\text{norm}}^2(A, A).$$

For a contraction algebra A ,

$$\dim T_A X_d = d(d-1) - \dim \text{Der}(A) + \dim HH^2(A) \quad (4.4)$$

$$= d(d-1) - d + \dim Z(A). \quad (4.5)$$

Proof. The tangent-space identification is Lemma 3.1. Normalized one-cochains satisfy $f(1) = 0$, so their dimension is $d(d-1)$; the kernel of their differential is $\text{Der}(A)$. Thus $\dim B_{\text{norm}}^2 = d(d-1) - \dim \text{Der}(A)$, giving the first equality. Since

$$\dim \text{Der}(A) = d - \dim Z(A) + \dim HH^1(A),$$

the second follows from Lemma 4.4. $\square$

Lemma 4.7 (Semisimple orbit dimension). *For split semisimple $S_{\text{GV}} = \prod_j M_j(\mathbb{C})^{\times n_j}$,*

$$\dim Z(S_{\text{GV}}) = \sum_j n_j, \quad \dim \text{Der}(S_{\text{GV}}) = d - \sum_j n_j,$$

and

$$\dim(G \cdot S_{\text{GV}}) = d(d-1) - d + \sum_j n_j. \quad (4.6)$$

Proof. The centre of S_{GV} has one dimension for each simple block, and there are $\sum_j n_j$ blocks. All derivations of a semisimple algebra over $\mathbb{C}$ are inner, so $\dim \text{Der}(S_{\text{GV}}) = d - \dim Z(S_{\text{GV}})$. The orbit dimension is then the case $S = S_{\text{GV}}$ of the computation in the proof of Corollary 3.5, the stabilizer group scheme being smooth in characteristic zero. $\square$

Combining (4.5) and (4.6),

$$\dim T_A X_d - \dim(G \cdot S_{\text{GV}}) = \dim Z(A) - \sum_j n_j. \quad (4.7)$$

Thus Theorem 3.4 and (4.3) reduce Theorem 4.1 to

$$\dim HH_0(A) = \sum_j n_j.$$

4.4 Universal marked cocenter on the Cartan cover

Fix one of the six primitive marked ADE pairs (Φ, p) . Here p is the retained vertex α_* of Section 2, and $m_p(\alpha) = c_*(\alpha)$. Let $\mathfrak{h}$ be the Cartan space, of rank r , and let

$$W_0 = \langle s_i : i \neq p \rangle, \quad H = \mathfrak{h}/W_0.$$

Write

$$\tilde{R} = \hat{\mathcal{O}}_{\mathfrak{h},0} \cong \mathbb{C}[[\lambda_1, \dots, \lambda_r]].$$

For a positive root $\alpha = \sum_i m_i(\alpha) \alpha_i$, put

$$N_p := \#\{\alpha \in \Phi^+ : m_p(\alpha) > 0\}. \quad (4.8)$$

Choose an orientation Q of the finite Dynkin diagram, and let $\bar{Q}$ be its double. Over $\tilde{R}$, set

$$\Pi_\lambda = \tilde{R}\bar{Q} / \left(\sum_{a \in Q_1} [a, a^*] - \sum_i \lambda_i e_i \right).$$

We use the convention of [13, Proposition 4.15] and write $(\lambda, \alpha) = \sum_i m_i(\alpha) \lambda_i$. Let

$$\mathcal{D}_p = e_p \Pi_\lambda e_p.$$

Theorem 4.8 (The marked cocenter on the Cartan cover). *For every primitive marked ADE pair (Φ, p) , there is an exact sequence*

$$0 \longrightarrow \tilde{R}^{N_p} \xrightarrow{\Psi_p} \tilde{R}^{N_p} \longrightarrow HH_0(\mathcal{D}_p/\tilde{R}) \longrightarrow 0 \quad (4.9)$$

such that

$$\det \Psi_p = \eta \Delta_p, \quad \Delta_p := \prod_{\substack{\alpha \in \Phi^+ \\ m_p(\alpha) > 0}} (\lambda, \alpha), \quad \eta \in \tilde{R}^\times. \quad (4.10)$$

Proof. We construct Ψ_p as multiplication by the central element on the cocenter of the Etingof–Rains extension.

The central extension. Take $\mu = \rho$, normalized by $\rho_i = (\rho, \alpha_i) = 1$. In the notation of Etingof–Rains [8], the universal parameters x_i satisfy $\sum_a [a, a^*] = \sum_i x_i e_i$, and z is adjoined centrally. The coefficient ring for the regular weight ρ is

$$K_\rho = \mathbb{C}[x_1 - \rho_1 z, \dots, x_r - \rho_r z].$$

The fibre at a parameter λ is the algebra Π_λ^ρ with relations

$$\sum_a [a, a^*] = \sum_i (\rho_i z + \lambda_i) e_i.$$

Via the coordinates $\lambda_i := x_i - \rho_i z$, completing K_ρ at its augmentation gives the formal power-series ring $\mathbb{C}[[\lambda_1, \dots, \lambda_r]]$, which is the completed Cartan parameter ring $\tilde{R}$. Here the λ_i are base coordinates and z is a central element of the algebra. The central extension is finite free over K_ρ by [8, Theorem 3.2(i)].

Define the marked corner of the completed universal central extension by

$$E_p := e_p(\Pi[z] \hat{\otimes}_{K_\rho} \tilde{R}) e_p, \quad \mathbf{M}_p := HH_0(E_p/\tilde{R}).$$

Since the Etingof–Rains central extension is finite free over K_ρ , the corner E_p is a finite $\tilde{R}$ -module. Therefore $\mathbf{M}_p$, as a quotient of E_p , is also finite over $\tilde{R}$.

The closed fibre of E_p is $e_p \Pi_0^\rho e_p$.

Generic rank. Etingof–Rains prove generic semisimplicity in [8, Theorem 3.2(v) and the display immediately following it]. That display decomposes the central extension into the ordinary deformed preprojective factors indexed by positive roots; the cited Crawley–Boevey theorem used there gives the simple module of dimension vector α . Hence, after passing to a splitting field $L/\text{Frac}(\tilde{R})$,

$$(E_p)_L \cong \prod_{m_p(\alpha) > 0} M_{m_p(\alpha)}(L). \quad (4.11)$$

Each matrix algebra contributes one dimension to its cocenter, so

$$\dim_L(M_p)_L = N_p. \quad (4.12)$$

The closed-fibre dimension. We need the special-fibre equality

$$\dim_{\mathbb{C}} HH_0(e_p \Pi_0^\rho e_p) = N_p. \quad (4.13)$$

We identify the cases supplied by the nodal-corner calculation of Etingof–Latour–Rains (ELR) [9].

At $\lambda = 0$, our normalization of ρ gives the central-extension relation used by ELR.

ELR call the branching vertex of a Dynkin diagram of type D or E the *nodal vertex*. Karmazyn’s six primitive flop presentations determine the retained vertex and the Levi Weyl subgroup W_0 . The retained vertices and Levi types follow from [13, Examples 4.22–4.27]. With that numbering, they are:

length	Dynkin type	retained vertex	W_0 in Karmazyn	deleted diagram	ELR status
1	A_1	unique	trivial	empty	immediate
2	D_4	$p = 4$	S_2^3	A_1^3	nodal D
3	E_6	$p = 6$	$S_2 \times S_3^2$	$A_1 \sqcup A_2^2$	nodal E_6
4	E_7	$p = 7$	$S_3 \times S_2 \times S_4$	$A_2 \sqcup A_1 \sqcup A_3$	nodal E_7
5	E_8	$p = 4$	$S_4 \times S_5$	$A_3 \sqcup A_4$	non-nodal
6	E_8	$p = 8$	$S_5 \times S_2 \times S_3$	$A_4 \sqcup A_1 \sqcup A_2$	nodal E_8

For a positive root α , ELR’s phrase “the root contains the nodal simple root” is equivalent to $m_p(\alpha) > 0$. Hence their root count is N_p .

For type D , ELR [9, Section 3, Lemma 3.4] prove the required *degreewise upper bounds* for the cocenter of the nodal corner. Summing those bounds gives

$$\dim_{\mathbb{C}} HH_0(e_p \Pi_0^\rho e_p) \leq N_p.$$

On the other hand, the generic decomposition (4.12) gives a finite $\tilde{R}$ -module M_p of generic rank N_p . For any finite module over the local domain $\tilde{R}$, the dimension of the closed fibre is at least the generic rank. Therefore the reverse inequality holds and equality follows.

For the nodal E_6, E_7, E_8 corners, ELR [9, Section 3, Case 2b] compute the same graded cocenter dimensions by exact Magma calculations and obtain equality with the positive-root count. These are the primitive flop lengths 3, 4, 6 in the table.

For A_1 , the closed corner $e_p \Pi_0^\rho e_p = \mathbb{C}[z]/(z)$ has one-dimensional cocenter. Section 4.5 computes dimension 104 for the coefficient-five E_8 vertex. This proves (4.13) in all six cases.

Freeness. Choose lifts of a basis of the special fibre (4.13). Since HH_0 is the cokernel of the commutator map, it commutes with arbitrary base change. In particular,

$$\mathbf{M}_p \otimes_{\tilde{R}} \mathbb{C} \cong HH_0(e_p \Pi_0^\rho e_p).$$

By (4.13), the right side has dimension N_p . Nakayama gives a surjection

$$\tilde{R}^{N_p} \twoheadrightarrow \mathbf{M}_p.$$

After tensoring with $\text{Frac}(\tilde{R})$, (4.12) shows that this map is an isomorphism. Its kernel is a submodule of a free module over the domain $\tilde{R}$, hence torsion-free. Since its generic fibre is zero, the kernel is zero. Therefore

$$\mathbf{M}_p \cong \tilde{R}^{N_p}. \quad (4.14)$$

Multiplication by the central element. Multiplication by z induces

$$\mu_z : \mathbf{M}_p \rightarrow \mathbf{M}_p.$$

Setting $z = 0$ recovers the ordinary deformed preprojective corner, and since z is central,

$$HH_0(\mathcal{D}_p/\tilde{R}) \cong \mathbf{M}_p/z\mathbf{M}_p = \text{coker } \mu_z. \quad (4.15)$$

On the root factor α , the decomposition displayed immediately after [8, Theorem 3.2(v)] gives

$$z|_{V_\alpha} = -\frac{(\lambda, \alpha)}{(\rho, \alpha)}.$$

Therefore the action of μ_z on the one-dimensional cocenter of that factor has the same eigenvalue, and over the fraction field

$$\det \mu_z = \prod_{m_p(\alpha) > 0} \left(-\frac{(\lambda, \alpha)}{(\rho, \alpha)} \right).$$

The denominators are nonzero complex constants, so

$$\det \mu_z = \eta \Delta_p, \quad \eta \in \tilde{R}^\times.$$

This determinant is nonzero; hence μ_z is injective over the domain $\tilde{R}$. Together with (4.14)–(4.15), this gives the required presentation and determinant. $\square$

4.5 The coefficient-five E_8 computation

Deleting the coefficient-five vertex of E_8 leaves $A_3 \times A_4$, so

$$N_p = |\Phi^+(E_8)| - |\Phi^+(A_3)| - |\Phi^+(A_4)| = 120 - 6 - 10 = 104. \quad (4.16)$$

The specialization of Karmazyn's presentation along the ρ -line, detailed in Appendix B.1, gives the closed marked corner $B_5 = e_p \Pi_0^\rho e_p$

$$B_5 = \frac{\mathbb{C}[z]\langle \beta, \gamma \rangle}{(\mathbf{P}, \mathbf{R}_4, \mathbf{R}_5)}, \quad (4.17)$$

where

$$\begin{aligned} P &= (\beta - 4z)(\beta - 5z)(\beta - 6z)(\beta - 7z), \\ R_4 &= (\gamma + \beta^2)^2 + \beta\gamma\beta + 5z^2(\gamma + \beta^2) - 4z^4, \\ R_5 &= \gamma\beta\gamma + \gamma^2\beta + \gamma\beta^3 + 5z^2\gamma\beta. \end{aligned}$$

Give B_5 the half-path grading

$$\deg z = \deg \beta = 1, \quad \deg \gamma = 2.$$

Proposition 4.9 (Exact modular calculation). *Let $h_m = \dim_{\mathbb{C}} HH_0(B_5)_m$. In degrees $0, \dots, 23$, the following sequence gives upper bounds for h_m :*

$$(1, 2, 4, 6, 7, 7, 7, 6, 6, 6, 6, 5, 5, 4, 4, 4, 4, 3, 3, 2, 2, 2, 2, 1).$$

The sum of these bounds is 99.

Proof. In degree m , use cyclic-word columns $z^a[w]_{\text{cyc}}$ with $a + \text{wt}(w) = m$. The rows are cyclic classes of homogeneous right multiples of P, R_4, R_5 . These span the relations in the cocenter: a two-sided multiple uRv has the same cyclic class as Rvu , and central powers of z can be moved to the front.

The relation matrices have integer entries. Exact Gaussian elimination over $\mathbb{F}_{1000003}$ gives the ranks recorded in Appendix B. A nonzero modular minor is also nonzero over $\mathbb{Q}$, so the modular ranks are lower bounds for the characteristic-zero ranks. Subtracting them from the column counts gives the stated upper bounds. $\square$

4.5.1 Tail bound and exact grading comparison

Write h for the Coxeter number. In the Etingof–Rains grading, arrows have degree 1 and z has degree 2. Their Hilbert-series formulas [8, Section 2.1, equation (1), and Theorem 3.2(iii)–(iv)] show that the ordinary preprojective algebra has top arrow degree $h - 2$, while the ρ -special central extension has top degree $2(h - 2)$. On the marked corner (z, β, γ) have ER degrees $(2, 2, 4)$. The finite computation uses exactly half this grading. For E_8 , $h = 30$, so the top ER degree is $2(h - 2) = 56$, which is computational half-degree 28. Setting $z = 0$ leaves the ordinary preprojective corner; because the E_8 graph is bipartite, its closed paths have even arrow length, and the top arrow-degree bound $h - 2 = 28$ makes it vanish in half-degree above 14. Since z is central,

$$HH_0(B_5)/(z) \cong HH_0(B_5/(z)).$$

Consequently multiplication by z is surjective

$$HH_0(B_5)_{m-1} \twoheadrightarrow HH_0(B_5)_m \quad (m > 14).$$

The finite calculation gives $\dim HH_0(B_5)_{23} \leq 1$. Hence every degree $24, \dots, 28$ has dimension at most one. Therefore

$$\dim HH_0(B_5) \leq 99 + 5 = 104.$$

The finite module M_p has generic rank $N_p = 104$ by (4.12), and its closed fibre is $HH_0(B_5)$. The closed-fibre dimension is at least the generic rank, so

$$\dim HH_0(B_5) \geq 104.$$

Therefore

$$\dim HH_0(B_5) = 104 = N_p. \quad (4.18)$$

This proves (4.13) in the remaining case.

4.6 Descent to the marked base

Write

$$\mathbb{H}_\Gamma = \mathcal{O}(\mathfrak{h}), \quad \mathbb{H}_\ell = \mathbb{H}_\Gamma^{W_0}, \quad R_H = \widehat{\mathbb{H}}_\ell, \quad \widetilde{R} = \widehat{\mathbb{H}}_\Gamma,$$

with completions at the graded origins. The extension $R_H \rightarrow \widetilde{R}$ is finite and faithfully flat. Indeed, W_0 is a reflection group, so both rings are regular local of the same dimension. A system of parameters of R_H is a system of parameters, hence a regular sequence, in the finite Cohen–Macaulay ring $\widetilde{R}$. Auslander–Buchsbaum implies that $\widetilde{R}$ is free over R_H ; its rank is positive.

4.6.1 The marked algebra on the Cartan cover

Let $\mathcal{A}_\Gamma$ be Karmazyn’s universal extended-Dynkin algebra and put $e_{0p} = e_0 + e_p$. Write $\mathcal{T}_{\Gamma,0p}$ for the two-summand generator on the Cartan partial resolution associated to this corner [13, Proposition 4.21], and $\mathcal{T}_\ell$ for the generator of Section 2.2. Completions below are taken at the central graded origins.

Proposition 4.10 (Comparison of the marked generators). *On the completed Cartan partial resolution there is an isomorphism*

$$\theta : \widehat{\mathcal{T}}_\ell \widehat{\otimes}_{R_H} \widetilde{R} \xrightarrow{\sim} \widehat{\mathcal{T}}_{\Gamma,0p} \quad (4.19)$$

preserving the structure-sheaf and retained-vertex summands. It induces an isomorphism

$$\widehat{\mathcal{A}}_\ell \widehat{\otimes}_{R_H} \widetilde{R} \xrightarrow{\sim} \widehat{e_{0p} \mathcal{A}_\Gamma e_{0p}} \quad (4.20)$$

sending e_0 to e_0 and e_p to e_p .

Proof. The Cartesian comparison in [13, Proposition 4.21] places the two generators on the same completed partial-resolution family over $\widetilde{R}$. This family is projective over its affine target, and source and target are flat over the complete local Noetherian base. On its central fibre, both generators restrict to

$$\mathcal{T}_0 = \mathcal{O} \oplus \mathcal{T}_{0,p},$$

where the summands are indexed by the affine and retained vertices.

Each summand of $\mathcal{T}_0$ has vanishing higher self-Ext. The uniqueness of its lift [14, Lemma 3.2] therefore gives isomorphisms between the two lifts of $\mathcal{O}$ and between the two lifts of $\mathcal{T}_{0,p}$. Their direct sum is θ , and it preserves both projectors. Taking opposite endomorphism algebras and applying Proposition 2.20 gives (4.20). $\square$

For an algebra Λ with idempotent e_0 , scalar extension commutes with the quotient $\Lambda/\Lambda e_0 \Lambda$, by right exactness. The corner identity

$$\frac{e_{0p} \mathcal{A}_\Gamma e_{0p}}{(e_{0p} \mathcal{A}_\Gamma e_{0p}) e_0 (e_{0p} \mathcal{A}_\Gamma e_{0p})} \cong e_p (\mathcal{A}_\Gamma / \mathcal{A}_\Gamma e_0 \mathcal{A}_\Gamma) e_p \quad (4.21)$$

then identifies the marked quotient upstairs. By [13, Proposition 4.15], $\mathcal{A}_\Gamma$ is the universal extended-Dynkin deformed preprojective algebra. Killing its affine vertex leaves the finite-Dynkin algebra Π_λ : the affine parameter relation eliminates t_0 , and the remaining parameters are $\lambda_i = t_i$. Thus

$$(\widehat{\mathcal{A}}_\ell / \widehat{\mathcal{A}}_\ell e_0 \widehat{\mathcal{A}}_\ell) \widehat{\otimes}_{R_H} \widetilde{R} \cong e_p \Pi_\lambda e_p = \mathcal{D}_p. \quad (4.22)$$

4.6.2 The cocenter on the marked base

Put

$$\mathcal{C}_H = \widehat{\mathcal{A}}_\ell / \widehat{\mathcal{A}}_\ell e_0 \widehat{\mathcal{A}}_\ell, \quad \mathbf{M}_H = HH_0(\mathcal{C}_H / R_H).$$

The completion is at the graded origin. Since $\widetilde{R}$ is finite free over R_H , tensoring a complete module with $\widetilde{R}$ gives a finite direct sum of copies of that module and is already complete. Consequently (4.22) gives

$$\mathcal{C}_H \otimes_{R_H} \widetilde{R} \cong \mathcal{D}_p. \quad (4.23)$$

Theorem 4.11 (The universal marked cocenter). *The algebra $\mathcal{C}_H$ is finite over R_H . Its cocenter has a free resolution*

$$0 \longrightarrow R_H^q \xrightarrow{\Theta} R_H^q \longrightarrow \mathbf{M}_H \longrightarrow 0 \quad (4.24)$$

with

$$\mathrm{Fitt}_0^{R_H}(\mathbf{M}_H) = (\det \Theta) = (P_1 \cdots P_\ell),$$

where $P_j = 0$ defines the reduced coefficient- j curve divisor.

Proof. The algebra $\mathcal{D}_p$ is finite over $\widetilde{R}$. Finiteness descends along the faithfully flat Cartan cover, so $\mathcal{C}_H$ and $\mathbf{M}_H$ are finite over R_H . The commutator-cokernel description gives

$$\mathbf{M}_H \otimes_{R_H} \widetilde{R} \cong HH_0(\mathcal{D}_p / \widetilde{R}). \quad (4.25)$$

Take a finite free surjection onto $\mathbf{M}_H$, with kernel K . After base change, Theorem 4.8 makes $K \otimes_{R_H} \widetilde{R}$ projective. Finite projectivity descends, and a finite projective module over the local ring R_H is free. Moreover $\mathbf{M}_H$ has generic rank zero, because its Cartan pullback has a square presentation with nonzero determinant. The two free ranks therefore agree, giving (4.24).

Fitting ideals commute with base change, so

$$\mathrm{Fitt}_0^{R_H}(\mathbf{M}_H) \widetilde{R} = (\Delta_p). \quad (4.26)$$

For $i \neq p$, the simple reflection s_i preserves m_p and sends every positive root other than α_i to a positive root. Thus W_0 permutes the factors of Δ_p without signs, and $\Delta_p \in \widetilde{R}^{W_0} = R_H$. Contracting the ideal in (4.26) along the faithfully flat extension gives

$$\mathrm{Fitt}_0^{R_H}(\mathbf{M}_H) = (\Delta_p). \quad (4.27)$$

Grouping the roots by marked coefficient and using Lemmas 2.5 and 2.6 gives

$$\Delta_p = \eta P_1 \cdots P_\ell, \quad \eta \in R_H^\times. \quad (4.28)$$

□

4.7 Restriction to the central arc

Let $R_1 = \mathbb{C}[[w]]$, and let $f_0 : \text{Spec } R_1 \rightarrow H$ be the classifying arc of the original flop. Set

$$\mathcal{C}_0 = \mathcal{C}_H \otimes_{R_H, f_0} R_1, \quad M_0 = HH_0(\mathcal{C}_0/R_1).$$

By Lemma 2.10 and Proposition 2.16, the arc meets the curve discriminant only at the origin, and $\text{ord}_w P_j(f_0(w)) = n_j$.

4.7.1 The contraction algebra on the arc

Let $\widehat{\mathcal{R}}_\ell$ be the completed universal target ring at its graded origin, and put

$$Z_0 = \widehat{\mathcal{R}}_\ell \otimes_{R_H} R_1, \quad \Lambda_0 = \widehat{\mathcal{A}}_\ell \otimes_{R_H} R_1.$$

Then $\mathcal{C}_0 = \Lambda_0/\Lambda_0 e_0 \Lambda_0$. Let $\mathfrak{m}$ denote the contraction maximal ideal of Z_0 .

Lemma 4.12 (The central-arc quotient). *The natural completion map $\mathcal{C}_0 \rightarrow \widehat{\mathcal{C}}_{0\mathfrak{m}}$ is an isomorphism, and*

$$\mathcal{C}_0 \cong \widehat{\mathcal{C}}_{0\mathfrak{m}} \cong A. \quad (4.29)$$

In particular,

$$HH_0(\mathcal{C}_0) \cong HH_0(A). \quad (4.30)$$

Proof. The classifying arc is nonconstant: the power series $g(w) = P_1(f_0(w))$ has positive finite order. Weierstrass division makes $\mathbb{C}[[w]]$ finite over $\mathbb{C}[[g(w)]]$. The latter subring is contained in the image of R_H , so R_1 is finite over R_H .

Consequently Z_0 is finite over the complete Noetherian local ring $\widehat{\mathcal{R}}_\ell$. Modulo the maximal ideal of that ring its fibre is $R_1/\mathfrak{m}_H R_1$, where $\mathfrak{m}_H$ is the maximal ideal of R_H . This fibre is local Artinian with residue field $\mathbb{C}$. Thus Z_0 is local and complete at $\mathfrak{m}$. The finite Z_0 -algebra Λ_0 and its quotient $\mathcal{C}_0$ are also complete.

The ordinary tensor products above therefore compute the completed pullback along the central arc. Proposition 2.21 identifies Λ_0 with the original basic NCCR, preserving e_0 . Taking the idempotent quotient gives $\mathcal{C}_0 \cong A$, and then (4.30) follows by taking cocenters. $\square$

4.7.2 The length calculation

Base change gives $M_0 = M_H \otimes_{R_H} R_1$. Write $\Theta_0 = f_0^* \Theta$. Its determinant is

$$\det \Theta_0 = \eta(w) \prod_{j=1}^{\ell} P_j(f_0(w)) \neq 0. \quad (4.31)$$

Thus the base change of (4.24) is exact:

$$0 \longrightarrow R_1^q \xrightarrow{\Theta_0} R_1^q \longrightarrow M_0 \longrightarrow 0. \quad (4.32)$$

Smith normal form over R_1 yields

$$\dim_{\mathbb{C}} M_0 = \text{length}_{R_1} M_0 = \text{ord}_w \det \Theta_0. \quad (4.33)$$

The relative and absolute cocenters have the same underlying $\mathbb{C}$ -vector space, since $r[a, b] = [ra, b]$ for central r . Combining Lemma 4.12, (4.31), and Proposition 2.16 gives

$$\dim_{\mathbb{C}} HH_0(A) = \sum_j \text{ord}_w P_j(f_0(w)) = \sum_j n_j. \quad (4.34)$$

Centre–cocenter duality then gives

$$\dim_{\mathbb{C}} Z(A) = \sum_j n_j. \quad (4.35)$$

Proof of Theorem 4.1. Equations (4.7) and (4.35) give

$$\dim T_A X_d = \dim(G \cdot S_{\text{GV}}). \quad (4.36)$$

The general existence theorem of Section 2, combined with Lemma 3.2, supplies

$$A \in \overline{G \cdot S_{\text{GV}}}. \quad (4.37)$$

Theorem 3.4 now shows that A is smooth, that $\overline{G \cdot S_{\text{GV}}}$ is its unique component, and that any split semisimple algebra whose orbit closure contains A is isomorphic to S_{GV} .

For any finite free formal deformation $\mathcal{D}$ of A with geometrically semisimple generic fibre, let S' be the split $\mathbb{C}$ -algebra of that geometric Wedderburn type. Lemma 3.2 gives $A \in \overline{G \cdot S'}$, and the orbit criterion yields

$$S' \cong S_{\text{GV}} = \prod_{j=1}^{\ell} M_j(\mathbb{C})^{\times n_j}. \quad (4.38)$$

□

4.8 Relative cocenters

Corollary 4.13. *Let $\mathcal{B}$ be a finite free $\mathbb{C}[[s]]$ -algebra with special fibre A and geometrically semisimple generic fibre. Then $HH_0(\mathcal{B}/\mathbb{C}[[s]])$ is free of rank $\sum_j n_j$.*

Proof. The relative cocenter is finite and commutes with arbitrary base change. Theorem 4.1 gives dimension $\sum_j n_j$ on the closed fibre and, after a splitting extension, on the generic fibre. Lemma 2.33 proves freeness. □

The marked two-parameter family. Pull back (4.24) along the morsification $F: \text{Spec } B \rightarrow H$, where $B = \mathbb{C}[[s, w]]$, and write

$$M_F = \text{coker}(\Theta_F : B^q \rightarrow B^q), \quad g = \det \Theta_F.$$

Its central determinant $g(0, w)$ is nonzero, of order $\sum_j n_j$. The adjugate identity gives $gM_F = 0$; Weierstrass preparation therefore makes M_F finite over $\mathbb{C}[[s]]$.

To check torsion-freeness, suppose $sv = \Theta_F y$ in B^q . Reducing modulo s and using the injectivity of Θ_0 gives $y = sy'$. Cancellation of s in the free module B^q then gives $v = \Theta_F y'$. Thus s is a non-zero-divisor on M_F , and M_F is free over $\mathbb{C}[[s]]$, of rank $\sum_j n_j$. This is the relative cocenter of the marked quotient obtained by scalar extension from $\mathcal{C}_H$.

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A Root enumeration for the primitive marked pairs

This appendix gives the finite calculation used in Lemma 2.5. Number the vertices in the diagrams of Figure 2 from left to right along the horizontal chain, and number the lower vertex last. Thus the retained vertices are 1, 2, 3, 3, 4, 3 for lengths 1, $\dots$, 6, respectively. This diagram numbering is used only for the enumeration; the table in Section 4.4 records Karmazyn's numbering.

Represent a root by its vector $v = (v_1, \dots, v_r)$ in the simple-root basis. If $i \sim h$ denotes adjacency in the diagram, then

$$s_i(v)_h = v_h \quad (h \neq i), \quad s_i(v)_i = -v_i + \sum_{h \sim i} v_h.$$

Starting from the simple roots, repeatedly apply all s_i and keep each new vector once. The process terminates because these are finite ADE root systems; by definition of the Weyl action, its output is the full root system. Retain the vectors with nonnegative coordinates to obtain Φ^+ .

For each marked coefficient $j > 0$, start with one vector satisfying $v_p = j$, and repeat the same procedure using only the reflections with $i \neq p$. Compare the resulting orbit with the complete set of positive roots of coefficient j . In all six cases they agree. The numbers of positive roots are

$$1, \quad 12, \quad 36, \quad 63, \quad 120, \quad 120,$$

and the non-Levi totals are

$$N_p = 1, \quad 9, \quad 29, \quad 53, \quad 104, \quad 106.$$

The individual orbit cardinalities are those of Lemma 2.5. The calculation also checks that the largest marked coefficient is the stated length.

The integer implementation is included in `tools/check_mathematics.py` in the source distribution. At each step it verifies

$$2 \sum_i v_i^2 - 2 \sum_{\{i,h\} \text{ an edge}} v_i v_h = 2.$$

All vector operations and comparisons are exact. Distinct coefficient orbits give disjoint sets of hyperplanes, since proportional roots in a reduced root system differ only by sign. This also verifies that the descended curve divisors in the lemma are distinct.

B The coefficient-five E_8 calculation

B.1 Specialization of the universal presentation

Karmazyn's length-five presentation has $W_0 \cong S_4 \times S_5$. On the ρ -line set the eight finite-node parameters equal to the central variable:

$$t_i = z, \quad i = 1, \dots, 8.$$

The affine relation in [13, Example 4.26] is

$$t_0 + 2t_1 + 3t_2 + 4t_3 + 5t_4 + 3t_5 + 2t_6 + 4t_7 + 6t_8 = 0.$$

The finite-node coefficients sum to 29, so

$$t_0 = -29z.$$

Karmazyn's invariant coordinates specialize to

$$t = \frac{4t_0 + 3t_1 + 2t_2 + t_3}{4} = -\frac{55}{2}z,$$

and the two arm-root sets, using the formulas in [13, Example 4.26], become

$$\begin{aligned} \{\tau_i^\delta\}_{i=1}^4 &= \left\{ \frac{1+2+3}{4}, \frac{1+2-1}{4}, \frac{1-2-1}{4}, -\frac{3+2+1}{4} \right\} z = \left\{ \frac{3}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{3}{2} \right\} z, \\ \{\tau_i^\delta\}_{i=1}^5 &= \left\{ \frac{1+2+3+4}{5}, \frac{1+2+3-1}{5}, \frac{1+2-2-1}{5}, \frac{1-3-2-1}{5}, \frac{-4-3-2-1}{5} \right\} z \\ &= \{2, 1, 0, -1, -2\}z. \end{aligned}$$

The δ -arm invariants are

$$\begin{aligned} T_2^\delta &= -\sigma_2(\tau_1^\delta, \dots, \tau_4^\delta) = \frac{5}{2}z^2, & T_1^\delta &= -\sigma_3(\tau_1^\delta, \dots, \tau_4^\delta) = 0, \\ T_0^\delta &= -\sigma_4(\tau_1^\delta, \dots, \tau_4^\delta) = -\frac{9}{16}z^4. \end{aligned}$$

Separately, the five-arm elementary-symmetric invariant substitutions give

$$T_3 = 5z^2, \quad T_2 = 0, \quad T_1 = -4z^4, \quad T_0 = 0,$$

because Karmazyn defines $(T_3, T_2, T_1, T_0) = (-\sigma_2, -\sigma_3, -\sigma_4, -\sigma_5)$ of the five τ_i 's. Finally $\delta = \beta + t/5 = \beta - \frac{11}{2}z$, so translating the four τ_i^δ -roots gives the four β -roots $4z, 5z, 6z, 7z$.

Karmazyn's universal length-five relations are

$$\begin{aligned} a\delta &= ta, & \delta a^* &= a^*t, \\ aa^* &= (t^4 - T_2^\delta t^2 - T_1^\delta t - T_0^\delta)e_0, \\ a^*a &= \delta^4 - T_2^\delta \delta^2 - T_1^\delta \delta - T_0^\delta e_4, \\ \beta - \delta + \frac{1}{5}te_4 &= 0, \\ \gamma\beta\gamma + \gamma^2\beta + \gamma\beta^3 &= -T_3\gamma\beta - T_2\gamma - T_0e_4, \\ (\gamma + \beta^2)^2 + \beta\gamma\beta &= -T_3(\gamma + \beta^2) - T_2\beta - T_1e_4, \end{aligned}$$

[13, Theorem 1.3 and Example 4.26]. In the quotient by the free vertex, $e_0 = a = a^* = 0$ and $e_4 = 1$. The retained-vertex relation $\delta = \beta + t/5$, together with $t = -55z/2$ and the displayed invariant specialization, gives

$$\begin{aligned} 0 &= \delta^4 - \frac{5}{2}z^2\delta^2 + \frac{9}{16}z^4 \\ &= \left(\delta - \frac{3}{2}z\right)\left(\delta - \frac{1}{2}z\right)\left(\delta + \frac{1}{2}z\right)\left(\delta + \frac{3}{2}z\right) \\ &= (\beta - 4z)(\beta - 5z)(\beta - 6z)(\beta - 7z). \end{aligned}$$

Substituting $(T_3, T_2, T_1, T_0) = (5z^2, 0, -4z^4, 0)$ in the last two universal relations and moving their right sides to the left gives exactly R_5 and R_4 in (4.17). Thus every relation in (4.17) is obtained from the cited universal presentation by the displayed substitutions.

B.2 Relation matrices and exact ranks

The C++17 checker enumerates cyclic-word columns and homogeneous relation rows, then performs sparse Gaussian elimination over $\mathbb{F}_{1000003}$. The Python controller verifies the archived source hash, compiles the checker, and compares each output row with the reference table. The source, outputs, and instructions are available at commit [5a2001f](#) through [the archived \$E_8\$ checker sources](#), [the recorded \$E_8\$ outputs](#), and [the \$E_8\$ computation and reproduction guide](#).

From the repository root, run

```
python3 reproducibility/cE8/code/verify_e8_archive.py
```

The table lists the degree, column count c_m , modular rank r_m , and characteristic-zero nullity bound $u_m = c_m - r_m$.

Degree m	Columns c_m	Rank r_m over $\mathbb{F}_{1000003}$	Nullity bound u_m
0	1	0	1
1	2	0	2
2	4	0	4
3	6	0	6
4	9	2	7
5	12	5	7
6	17	10	7
7	22	16	6
8	30	24	6
9	40	34	6
10	55	49	6
11	74	69	5
12	105	100	5
13	146	142	4
14	210	206	4
15	304	300	4
16	447	443	4
17	658	655	3
18	987	984	3
19	1480	1478	2
20	2246	2244	2
21	3416	3414	2
22	5227	5225	2
23	8014	8013	1

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